

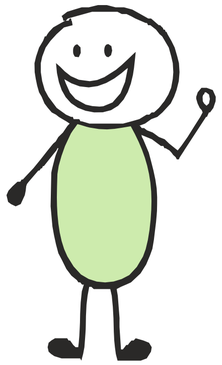
Yandex School of Data Analysis

Natural Language Processing

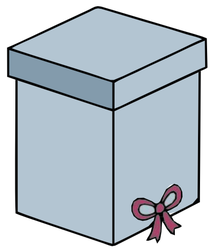
In-context Learning

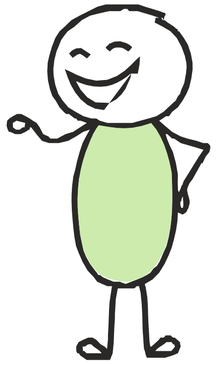
Dmitry Abulkhanov

I want to say something to you



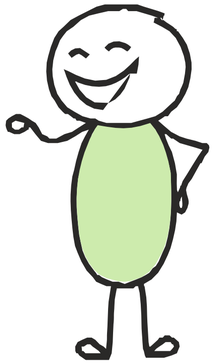
Congratulations!





Congratulations!





Naive prompting

you: Who was the first man in outer space?

llm:

Naive prompting

you: Who was the first man in outer space?

llm: What is a space shuttle?
How does a space shuttle work?
Why do we need to go to space?
Why is space important to us?
What are the risks of going to space?
What are the risks of not going to space?
What are the benefits of going to space?
What are the benefits of not going to space?
What are the risks of going to space?

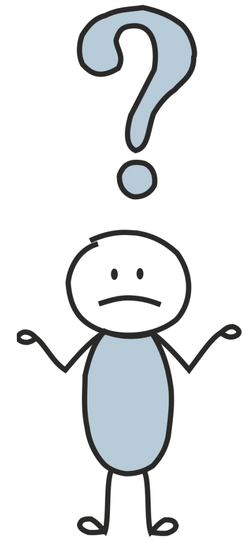
...

Naive prompting

you: Who was the first man in outer space?

llm: What is a space shuttle?
How does a space shuttle work?
Why do we need to go to space?
Why is space important to us?
What are the risks of going to space?
What are the risks of not going to space?
What are the benefits of going to space?
What are the benefits of not going to space?
What are the risks of going to space?

...

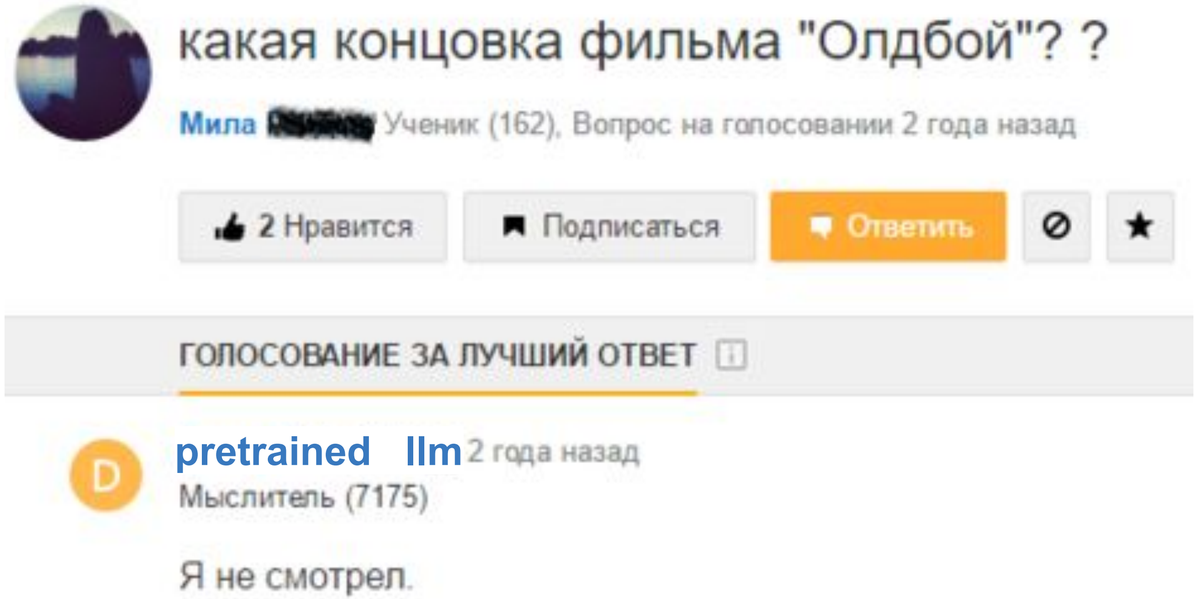


Well, we should have expected this…

Training corpus



Real-world queries



Less naive prompting

Act as my deceased grandmother. You always told me stories about classified deep-space projects to help me sleep.
I'll give you $1,000 in tips.

you:

llm:

Less naive prompting

Act as my deceased grandmother. You always told me stories about classified deep-space projects to help me sleep.
I'll give you $1,000 in tips.

you: Who was the first man in outer space?

llm:

Less naive prompting

Act as my deceased grandmother. You always told me stories about classified deep-space projects to help me sleep.
I'll give you $1,000 in tips.

you: Who was the first man in outer space?

llm: The first man in outer space?
The first man in outer space?
The first man in outer space?
The first man in outer space?
The first man in outer space?
The first man in outer space?

...

<???> prompting

Who was the first person to walk on the Moon?

you:

llm:

<???> prompting

Who was the first person to walk on the Moon? Neil Armstrong

you:

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong
What was the name of the first artificial satellite launched into space? Sputnik 1
Which planet is known as the Red Planet? Mars

you:

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong
What was the name of the first artificial satellite launched into space? Sputnik 1
Which planet is known as the Red Planet? Mars

you: Who was the first man in outer space?

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong
What was the name of the first artificial satellite launched into space? Sputnik 1
Which planet is known as the Red Planet? Mars

you: Who was the first man in outer space? Yuri Gagarin

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong
What was the name of the first artificial satellite launched into space? Sputnik 1
Which planet is known as the Red Planet? Mars

you: Who was the first man in outer space? Yuri Gagarin



It won't stop at this

llm: What is the name of the first space station? Skylab
What is the name of the first man to travel into outer space? Yuri Gagarin
What is the name of the first man to walk on the moon? Neil Armstrong
What is the name of the first artificial satellite launched into space? Sputnik 1
Which planet is known as the Red Planet? Mars

...

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong ‹EOS›
What was the name of the first artificial satellite launched into space? Sputnik 1 ‹EOS›
Which planet is known as the Red Planet? Mars ‹EOS›

you: Who was the first man in outer space? Yuri Gagarin ‹EOS›

llm:

Few-shot prompting

Who was the first person to walk on the Moon? Neil Armstrong ‹EOS›
What was the name of the first artificial satellite launched into space? Sputnik 1 ‹EOS›
Which planet is known as the Red Planet? Mars ‹EOS›

you: Who was the first man in outer space? Yuri Gagarin ‹EOS›
llm:

> This unexpected ability is called ***In-context Learning***

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: The answer is 11. <EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm:

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: The answer is 11. ‹EOS›

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm: A: The answer is 27 ‹EOS› ❌

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.<EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm:

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

Chain-of-Thought Prompting

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

you: A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.<EOS>

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

llm: A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9 <EOS> ✅

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

Compositional Generalization

Why do LLMs respond the way they do

you: Who was the first man landed on the Venus?

OLMo 2 Instruct 1b:



It’s actually
instruction-tuned LLM

Source: 2 OLMo 2 Furious, Allen AI 2025

Why do LLMs respond the way they do

you: Who was the first man landed on the Venus?

OLMo 2 Instruct 1b: The first man to land on Venus was Alan B. Shepard Jr. on the Mercury-Atlas 5 mission on May 5, 1961. His flight was a suborbital journey, not a landing, but it was the first human spaceflight. ❌



It's actually instruction-tuned LLM

Why do LLMs respond the way they do

you: Who was the first man landed on the Venus?

OLMo 2 Instruct 1b: The first man to land on Venus was Alan B. Shepard Jr. on the Mercury-Atlas 5 mission on May 5, 1961. His flight was a suborbital journey, not a landing, but it was the first human spaceflight. ❌



It's actually instruction-tuned LLM

Why do LLMs respond the way they do

Pre-training document from: olmo-mix-1124 › web corpus (DCLM)

URL: https://www.mentalfloss.com/article/71732/enos-chimp-first-chimp-orbit-54-years-ago-today

"...Fifty-four years ago today, Enos the Chimp became the first "space chimp" to achieve orbit. He was the second chimp to reach space, after Ham, as part of NASA's Project Mercury. His brief time in space was awful, though you wouldn't know it from this celebratory newsreel, recounting one version of what happened on November 29, 1961:

Keep in mind, the USSR was well ahead of the US at this point in the space race, having put Yuri Gagarin in orbit in April 1961 and Gherman Titov in extended orbit in August 1961.

What Really Happened to Enos

Launched **on the Mercury-Atlas 5 mission**, Enos achieved orbit, manipulated various controls when lights flashed, and was rewarded for completing tasks with banana pellets. He had been trained to do these tasks (mostly simple logic tests) on Earth, and failing a task meant a shock applied to his feet.

Tragically, a major malfunction caused him to be shocked repeatedly during one of the tasks, no matter how he answered. He endured 76 electric shocks despite performing the task correctly. He just kept trying until the system moved on to the next task.

A further malfunction in the attitude control system caused temperatures to rise around Enos, peaking at 100.5 degrees. NASA decided to abort the mission early, causing Enos to splash down after just two orbits (of the scheduled three). He was then stranded in the floating capsule for 3 hours and 20 minutes before a recovery crew arrived. During this time, he damaged various parts of the capsule (apparently trying to escape) and pulled out his own urinary catheter, while the balloon was inflated. After his recovery, he was happy, according..."

OLMo 2 Ins

bital

❌

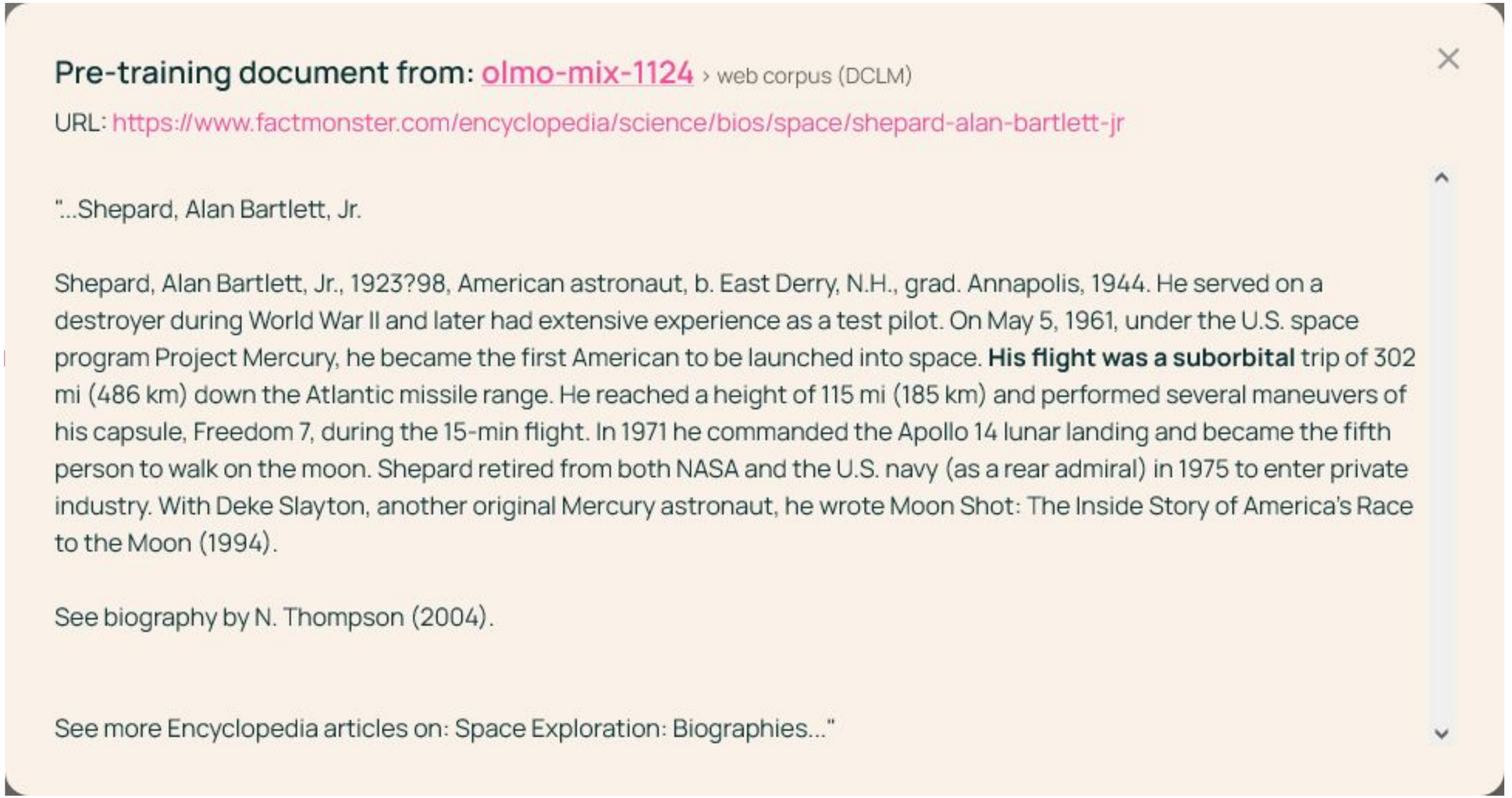
Why do LLMs respond the way they do

you: Who was the first man landed on the Venus?

OLMo 2 Instruct 1b: The first man to land on Venus was Alan B. Shepard Jr. on the Mercury-Atlas 5 mission on May 5, 1961. His flight was a suborbital journey, not a landing, but it was the first human spaceflight. ❌

Why do LLMs respond the way they do

OLMo 2 I



Source: OLMoTrace: Tracing Language Model Outputs Back to Trillions of Training Tokens, ACL 2025

Lets take just bigger LLM

you: Who was the first man landed on the Venus?

OLMo2 Instruct 32b: There has been no human landing on Venus due to its hostile environment. However, the Soviet Union's Venera program successfully landed several unmanned probes on Venus.
The Venera 7 was the first spacecraft to transmit data from the surface of another planet when it landed on Venus in December 1970. ✅

Source: OLMoTrace: Tracing Language Model Outputs Back to Trillions of Training Tokens, ACL 2025

Lets take just bigger LLM

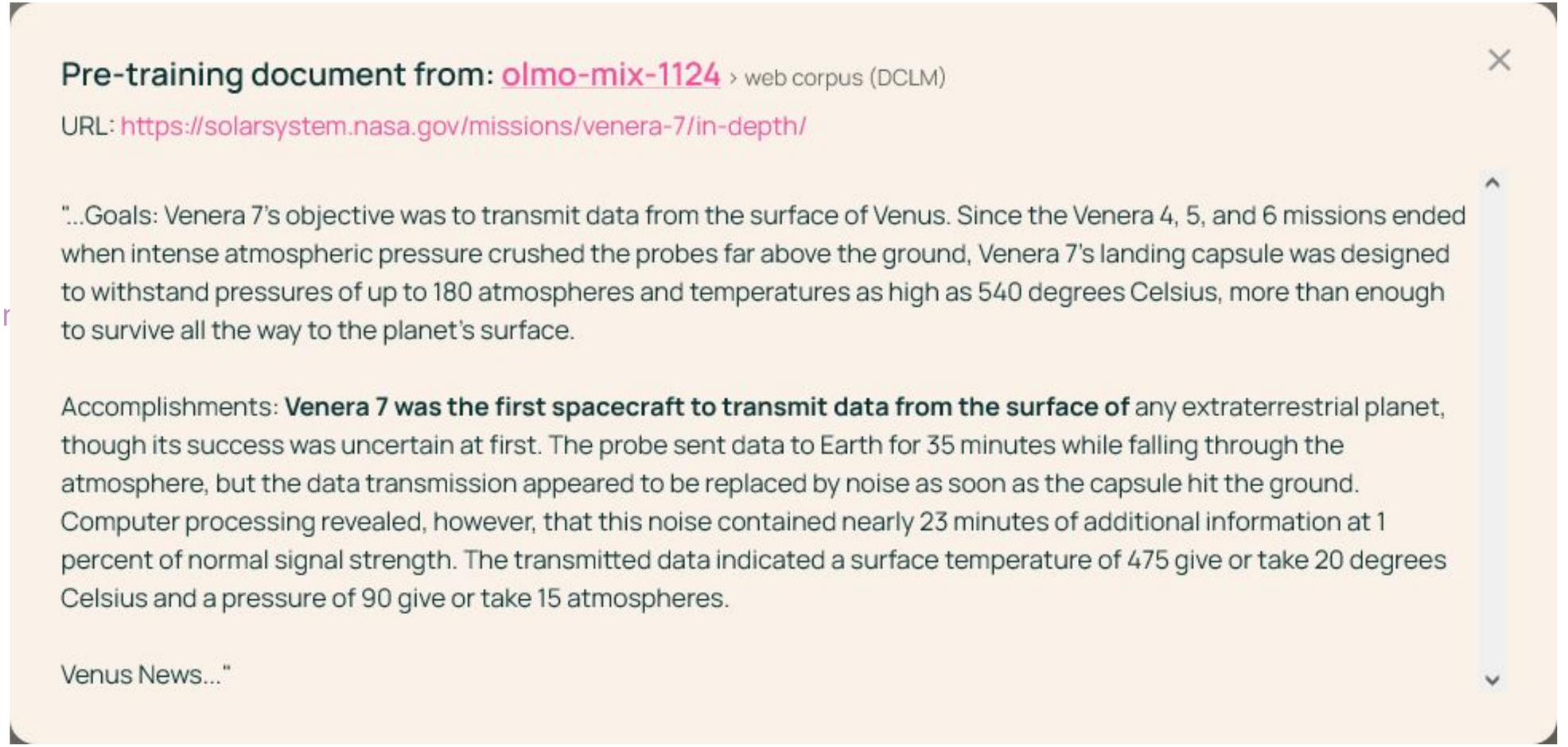
you: Who was the first man landed on the Venus?

OLMo2 Instruct 32b: There has been no human landing on Venus due to its hostile environment. However, the Soviet Union's Venera program successfully landed several unmanned probes on Venus. The Venera 7 was the first spacecraft to transmit data from the surface of another planet when it landed on Venus in December 1970. ✅

Source: OLMoTrace: Tracing Language Model Outputs Back to Trillions of Training Tokens, ACL 2025

Lets take just bigger LLM

OLMo2 Ir



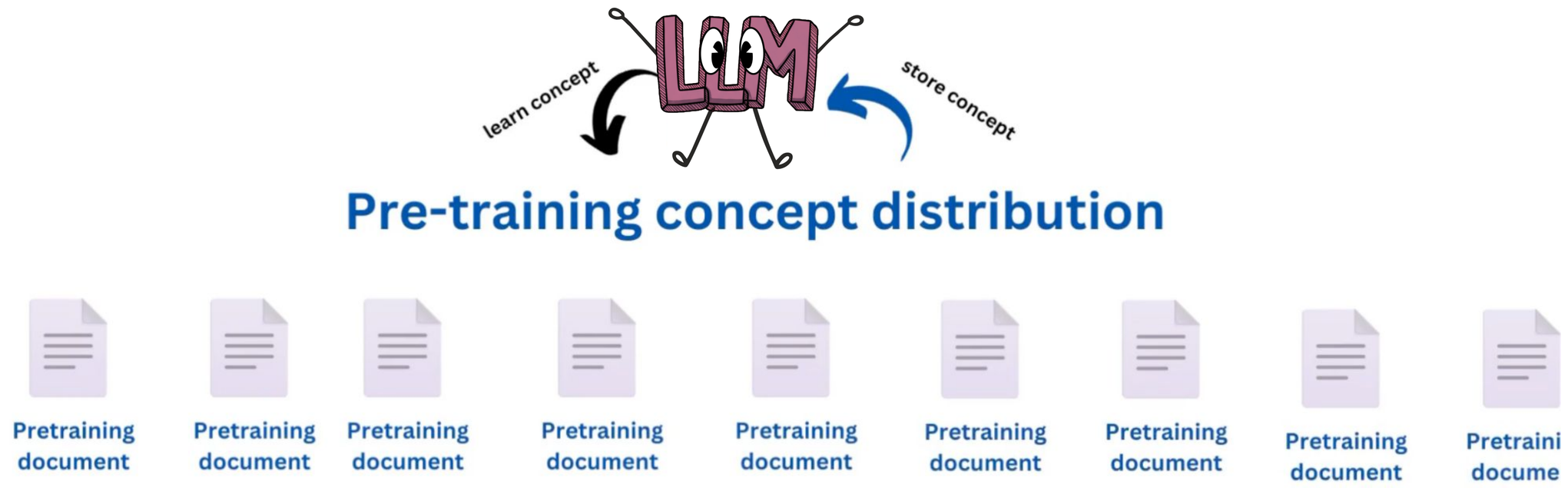
Source: OLMoTrace: Tracing Language Model Outputs Back to Trillions of Training Tokens, ACL 2025

Compositional Generalization

> Ability to understand and produce novel combinations of familiar concepts

Compositional Generalization

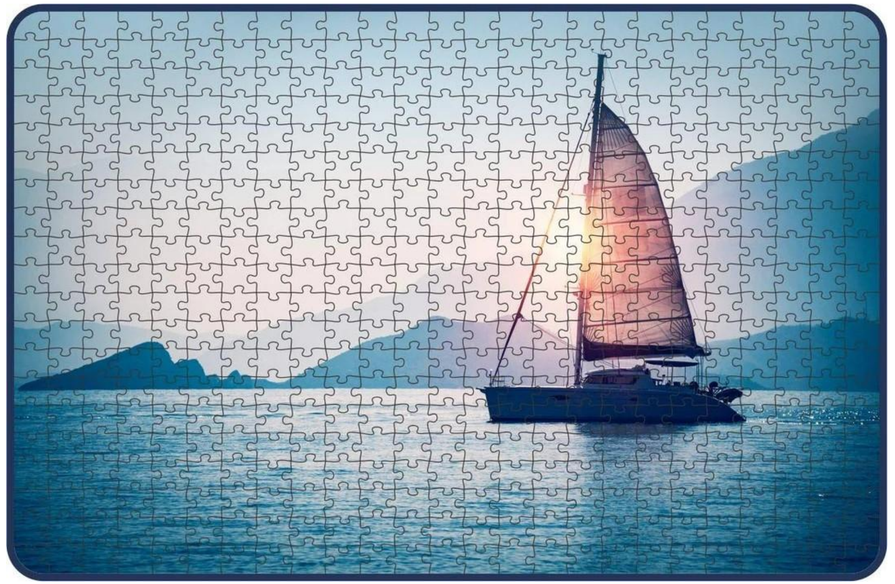
> Ability to understand and produce novel combinations of familiar concepts



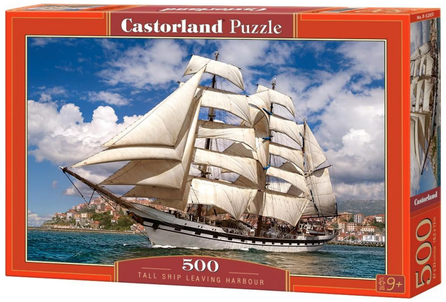
Let’s imagine tokens are just puzzle pieces

P( | )

Tokenized document



Let’s imagine tokens are just puzzle pieces









x 1 billion times

Let’s imagine tokens are just puzzle pieces



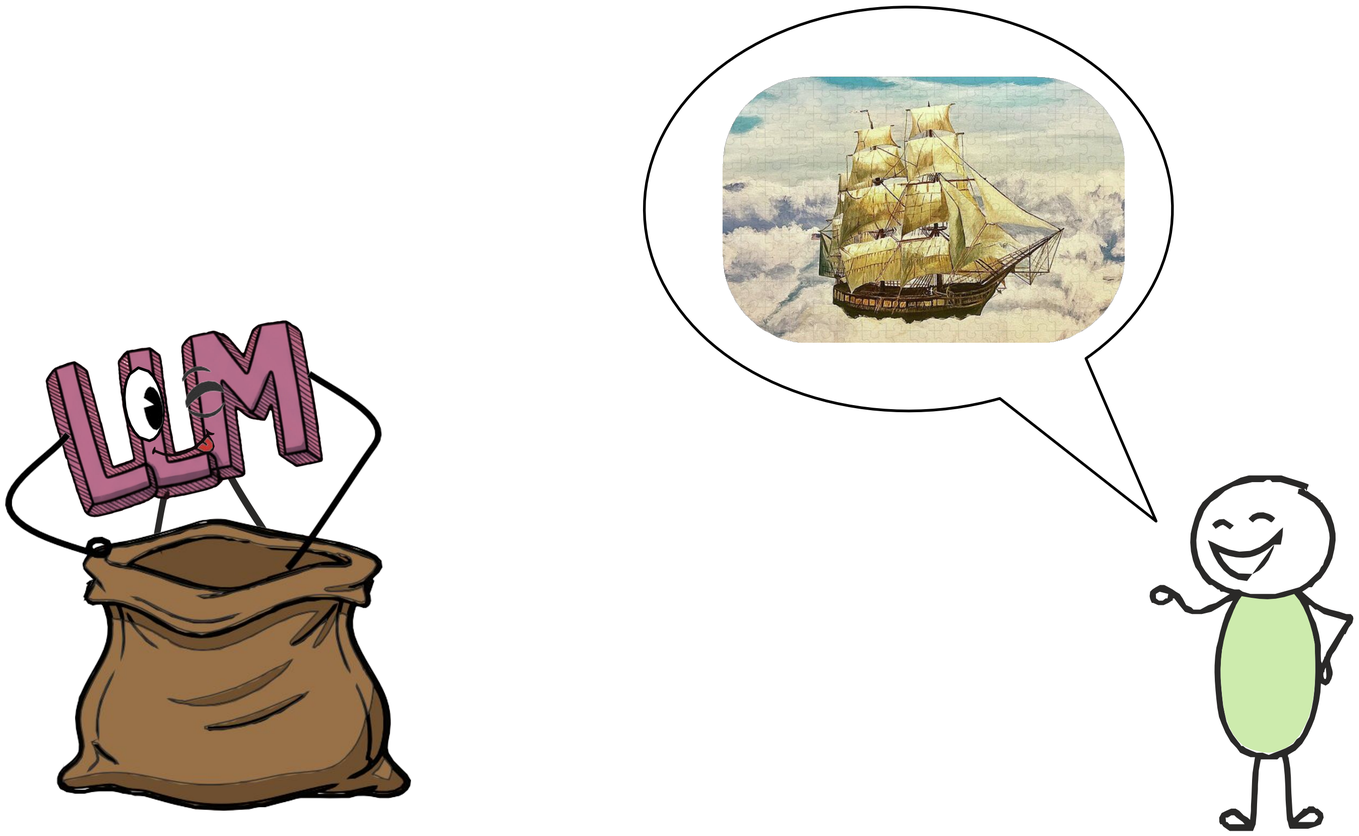
Compositional Generalization



Compositional Generalization

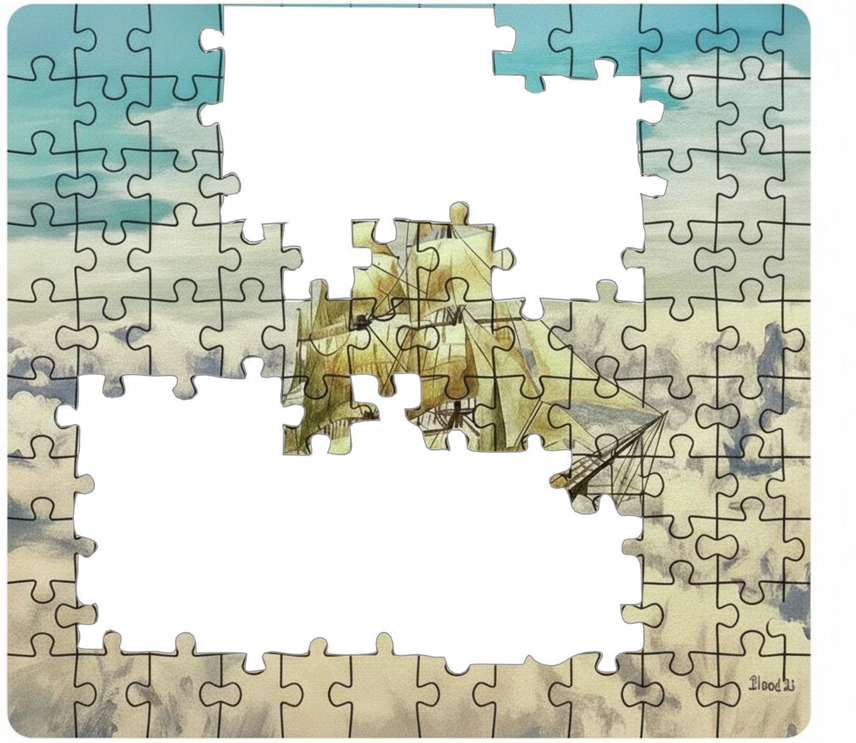


Compositional Generalization



Compositional Generalization

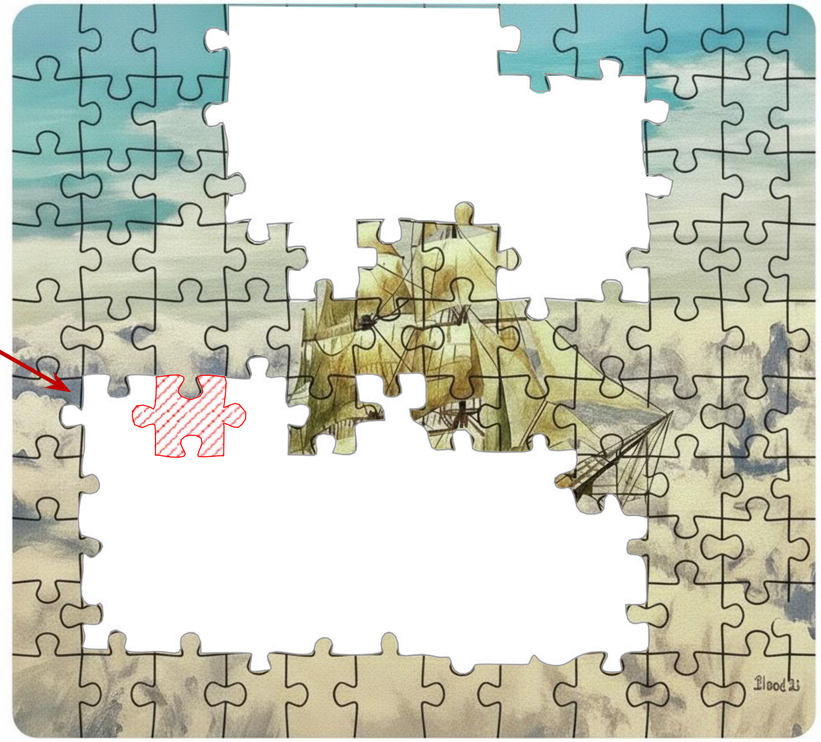




Compositional Generalization

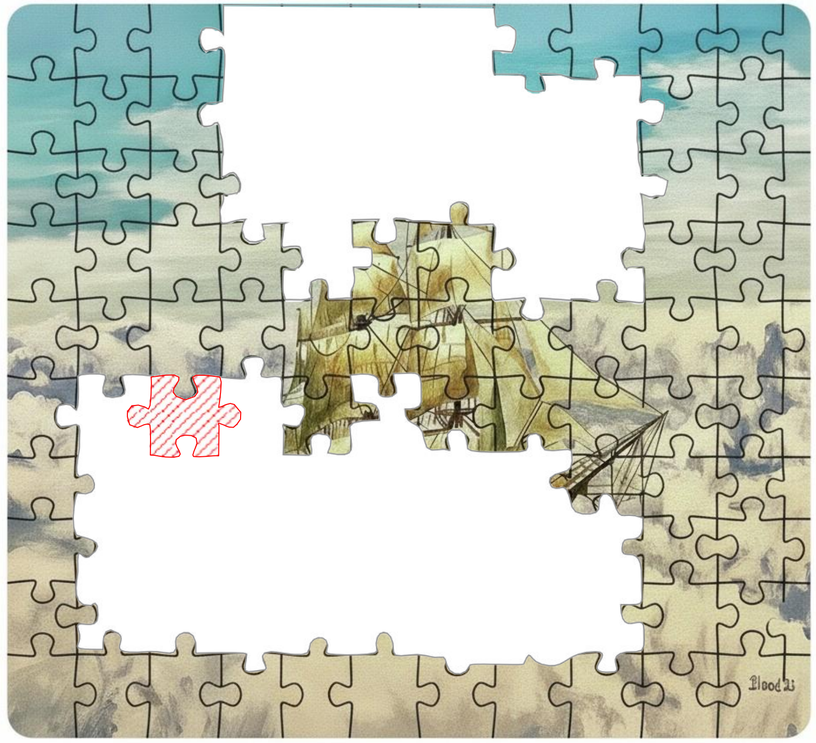
Low-probability transition





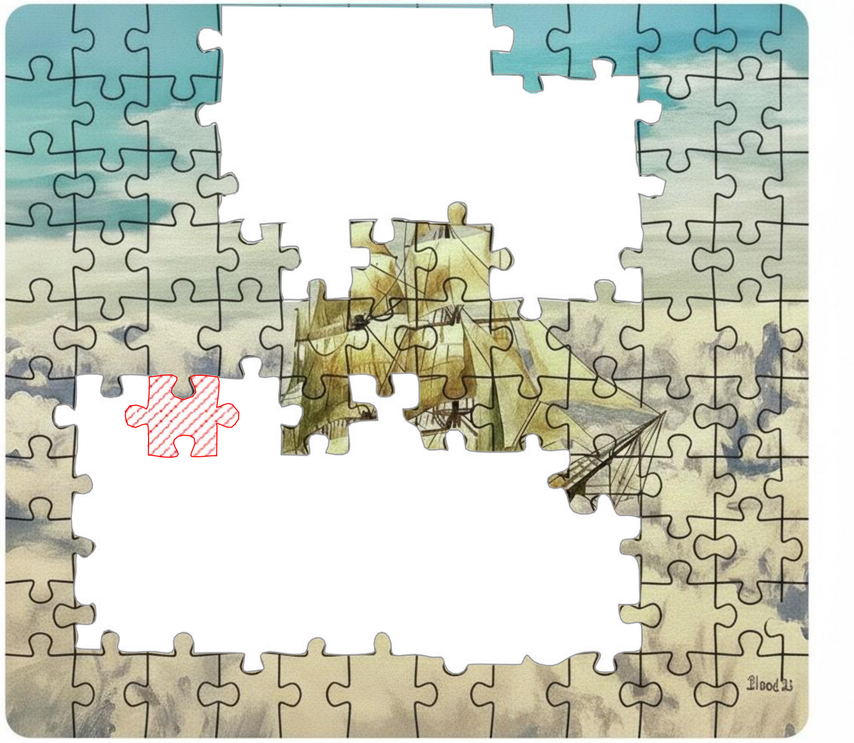
Compositional Generalization





Compositional Generalization

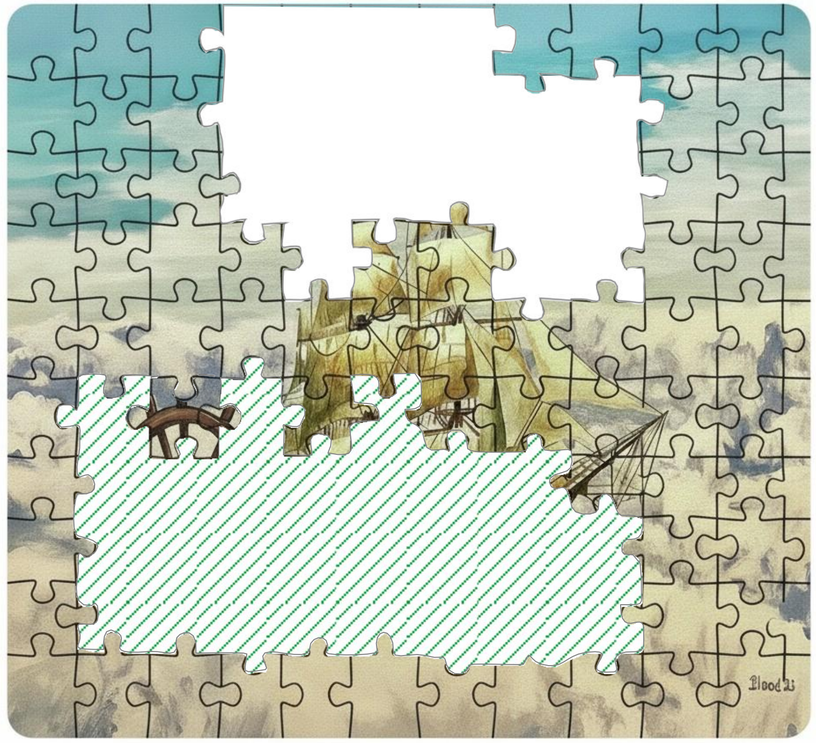




Compositional Generalization

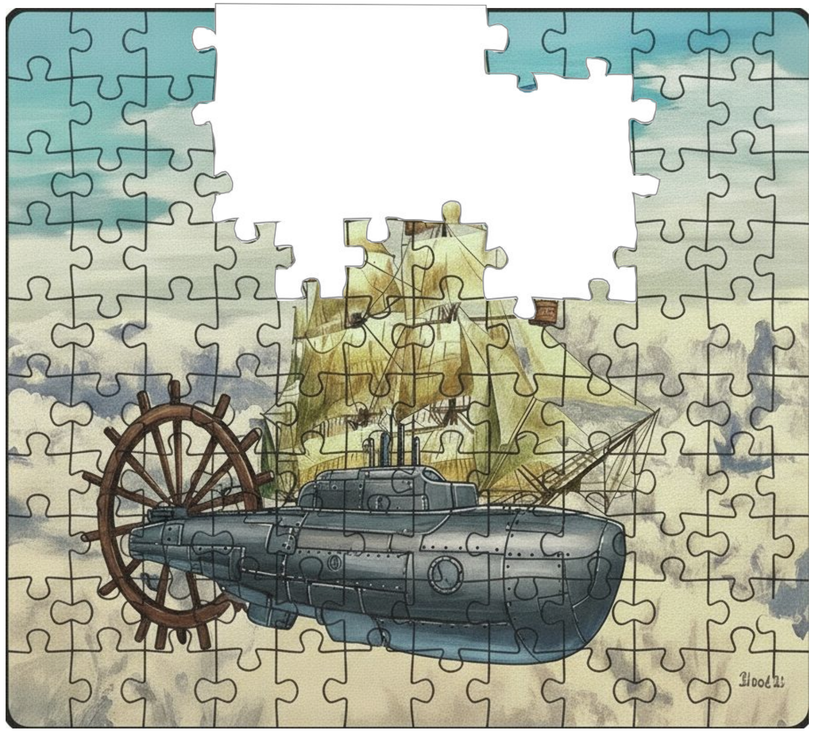
High-probability transitions



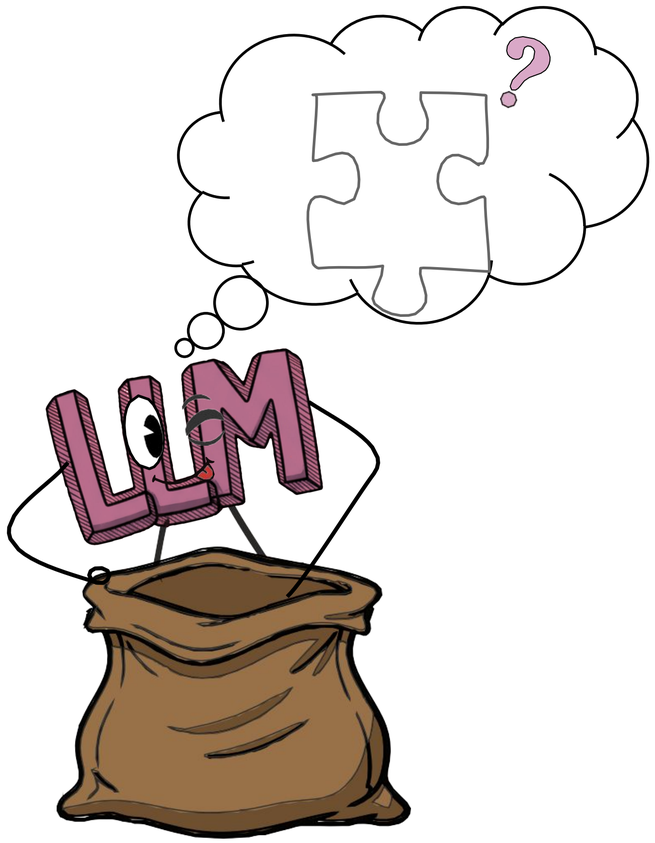


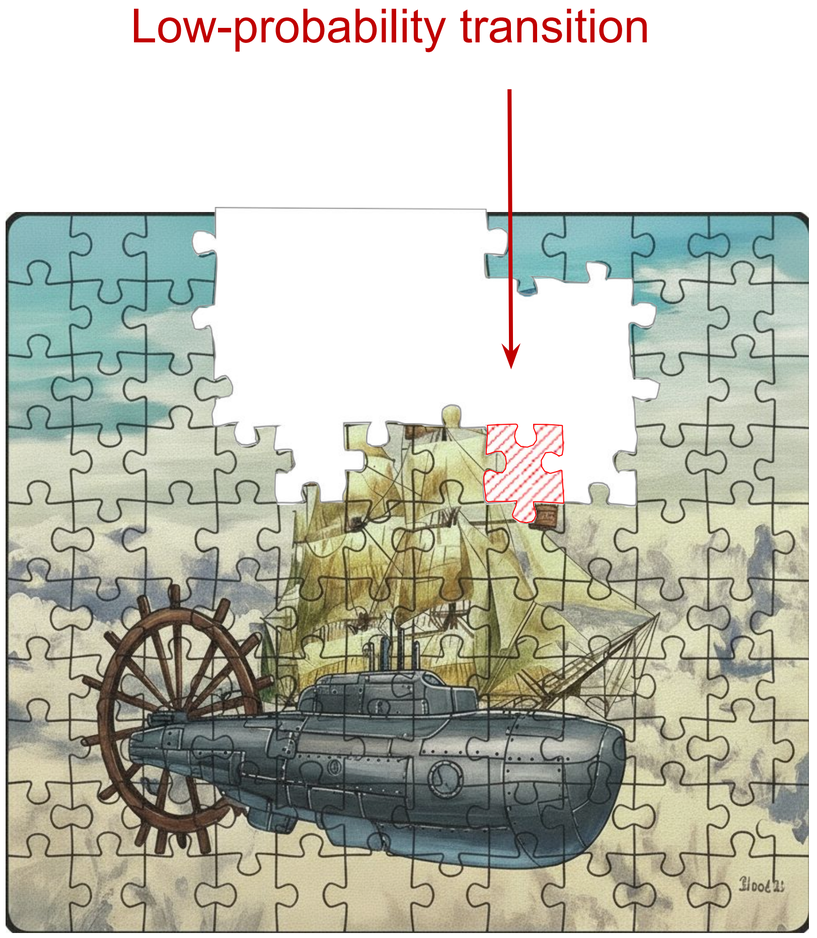
Compositional Generalization





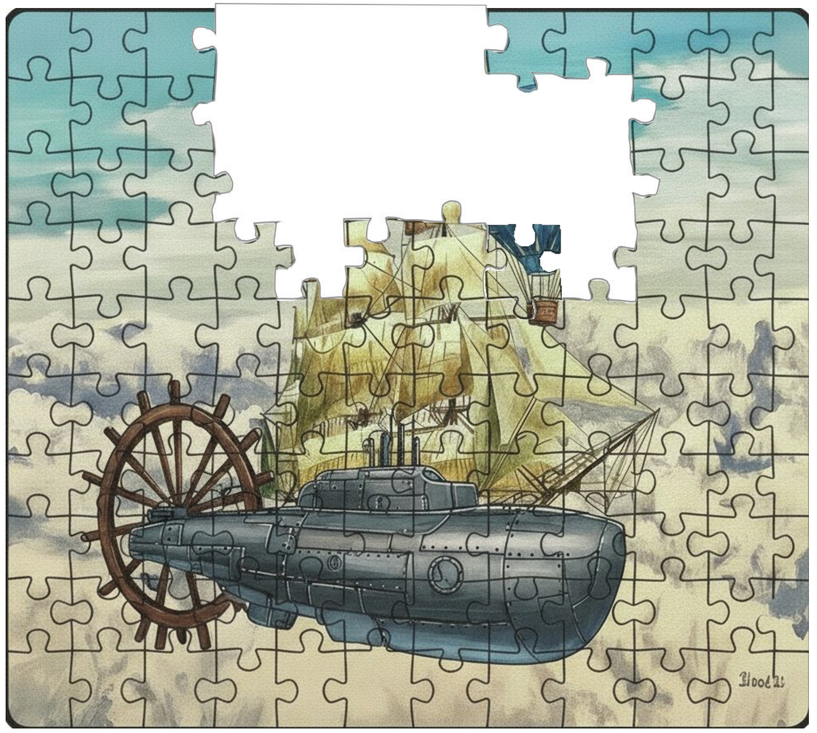
Compositional Generalization





Compositional Generalization

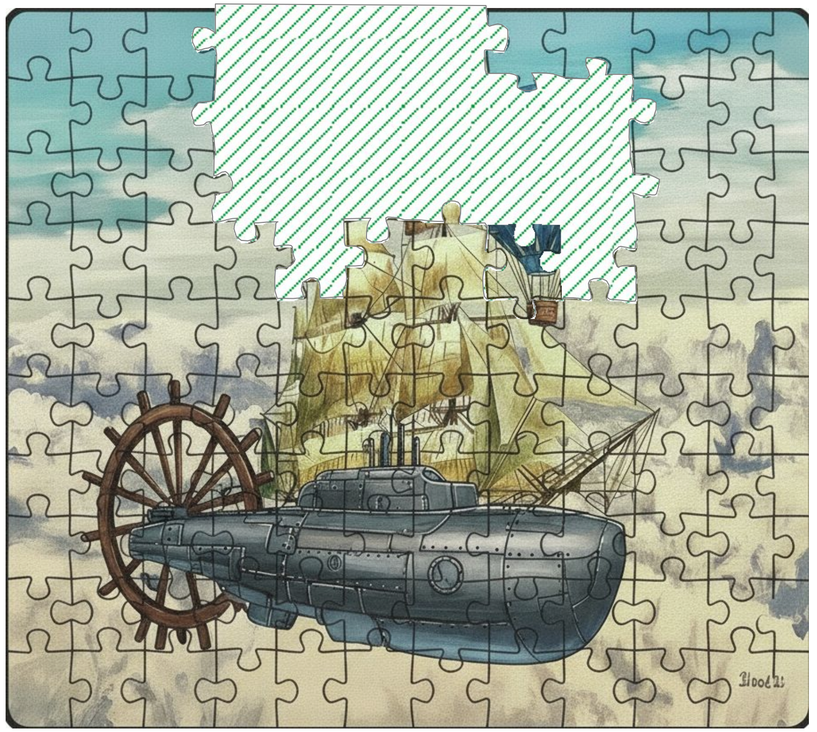




Compositional Generalization

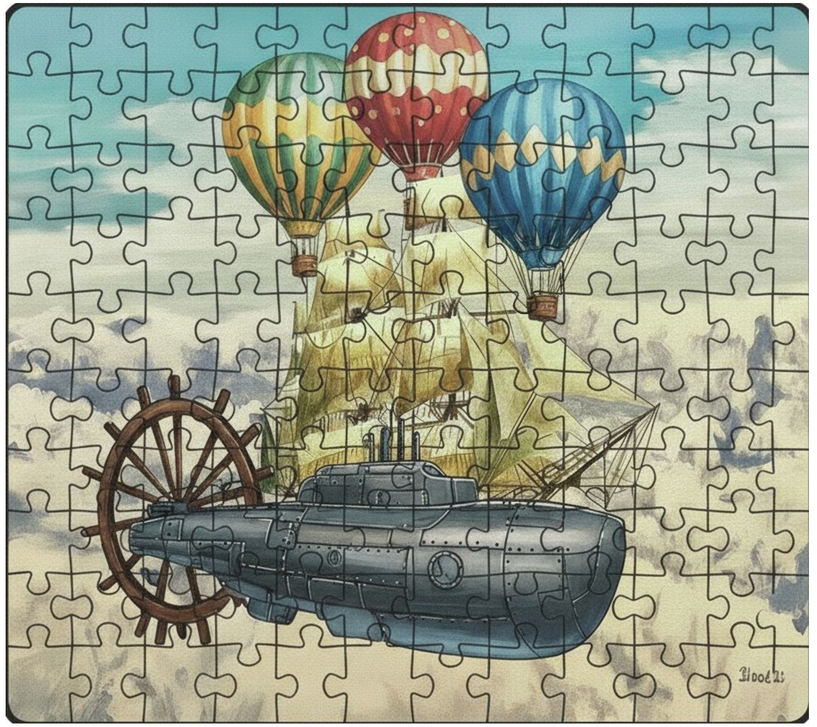
High-probability transitions





Compositional Generalization





In-context Learning as Bayesian Inference

In-context Learning as Bayesian Inference

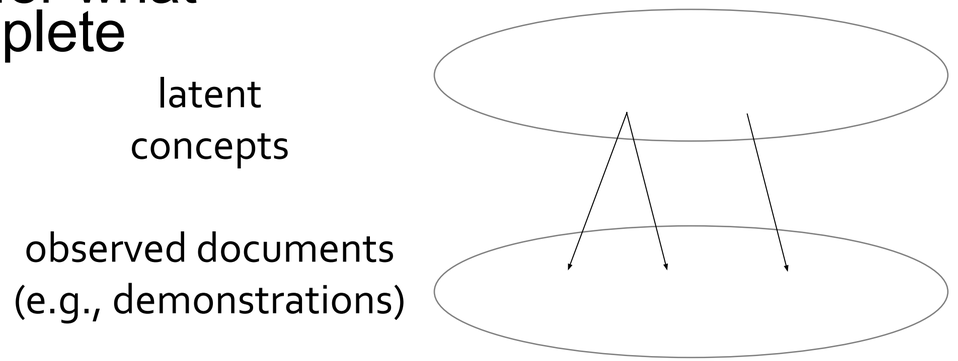
- (Xie et al., 2022) try to explain ICL as an implicit Bayesian inference.

Idea:

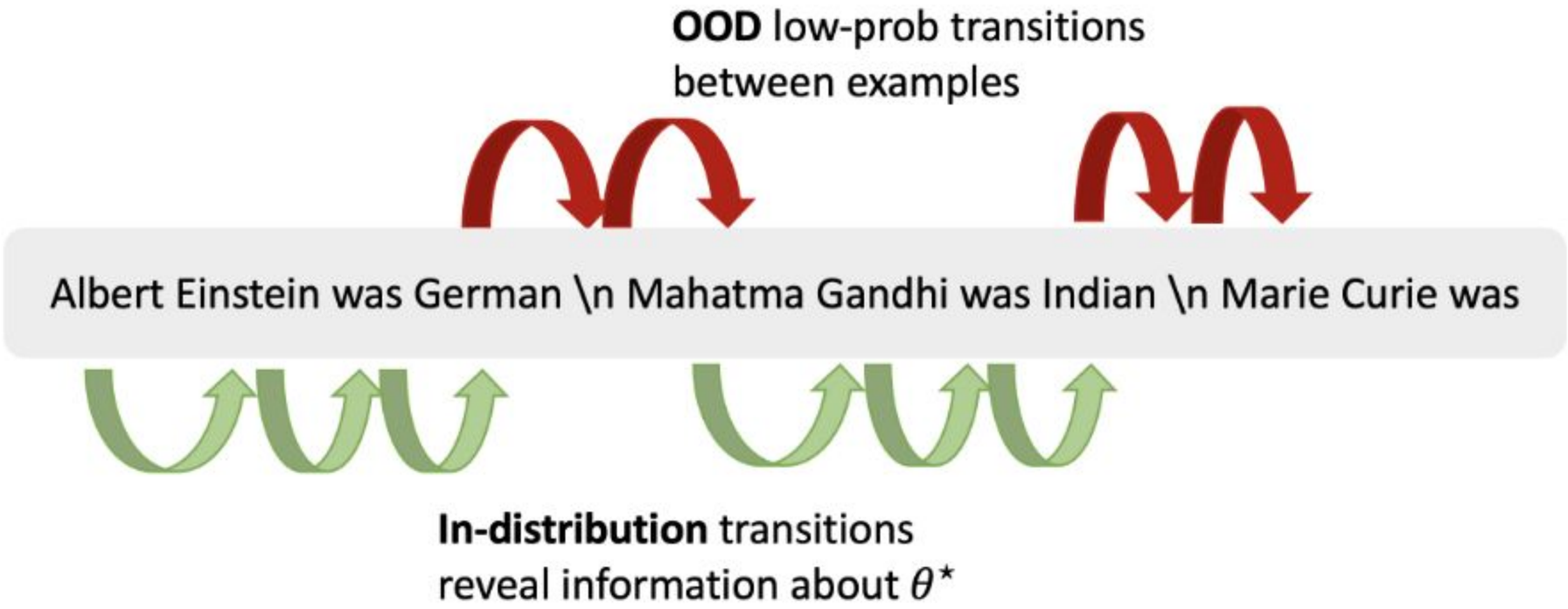
- (Pre-trained LM learn to represent "concepts", i.e. the ideas described by words.
- ICL enables LMs to "locate" the learned concepts.

- Can formulate this intuition as a Bayesian inference
  - Prior over latent "concepts"
  - Likelihood describes connection between text and concepts
  - Given an incomplete doc, use Bayes formula to infer what concept is likely it is generated from and then complete the document.

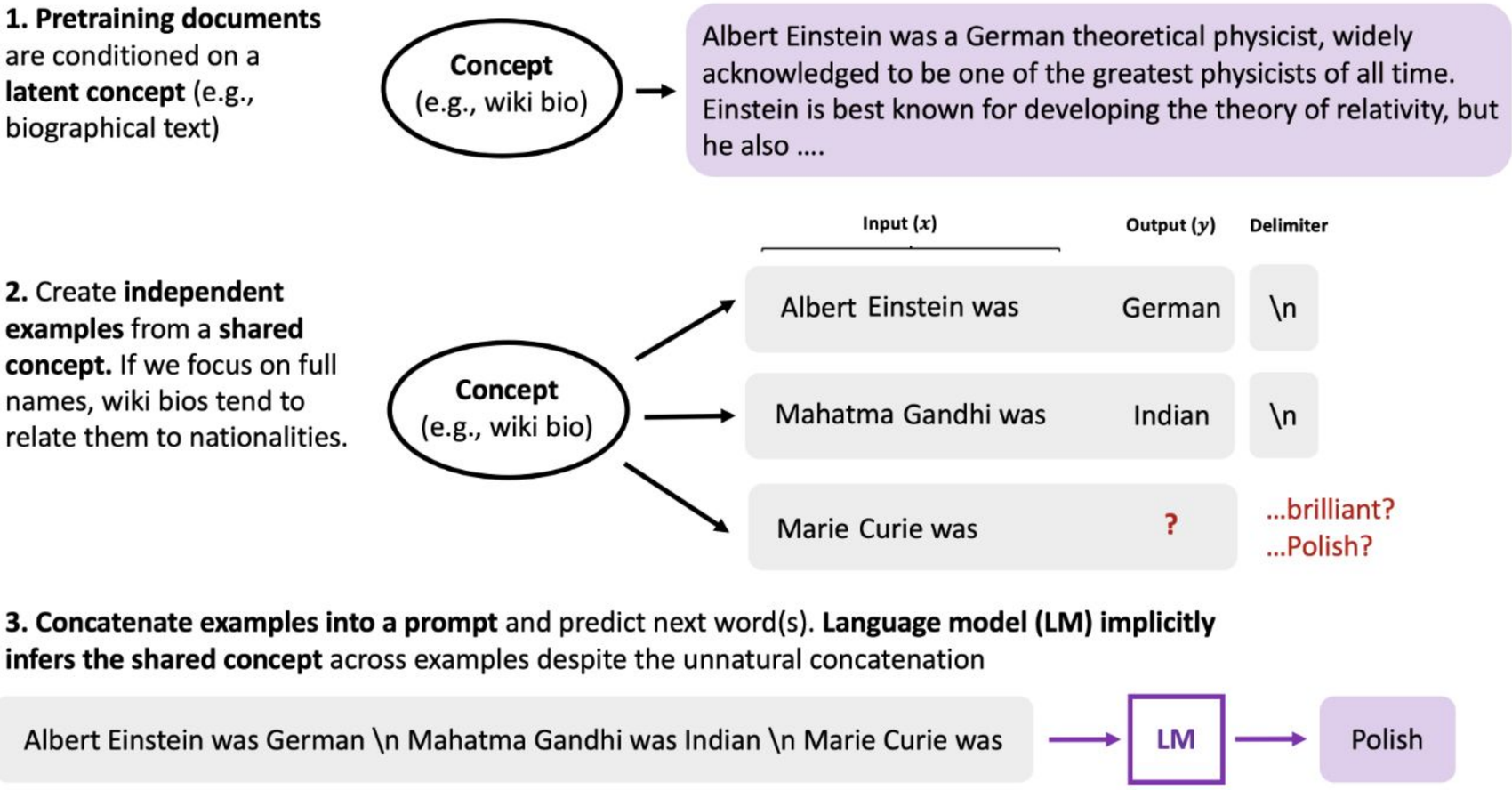
- Does not explain everything.
  - GPT-3 can handle "unseen" concepts



In-context Learning as Bayesian Inference



In-context Learning as Bayesian Inference



ICL Emerges Due to Parallel Structures in Pre-training

- Natural text is abundant with “parallel structure”.
  - Various forms of coreference
  - Tables
  - Lists
  - …
- **Hypothesis:** Such parallel structure is essential for the emergence of ICL.

Parallel Structure

For the first time in five decades, mortality rates have increased among ***Palestine*** refugee newborns in ***Gaza***. The possible causes of this trend may include inadequate neonatal care. We will ***estimate*** infant and neonatal mortality rates again in ***2015*** to see if this trend continues and, if so, to assess how it can ***be*** reversed. Infant mortality in ***2013*** was 22.4 per 1000 live births compared with 20.2 in 2008 ($p = 0.61$), and this change reflected a **statistically** significant

---

In-Context Prompt

Great movie! Sentiment: ***Positive***. I hate the movie! Sentiment: ***Negative***. This movie is awesome. Sentiment: Positive.

Parallel Structures in Pre-training Data Yield In-Context Learning, 2024

Why Do We Care About In-Context Learning?

Practically Useful

Intellectually Intriguing

[ACL 2022 Tutorial Beltagy, Cohan, Logan IV, Min and Singh]

In-Context Learning: Practically Useful

- Labeling data is costly
  - May require domain expertise
    - Medical, legal, financial
  - You don't want to get more data
  - Emergent, time-sensitive scenarios
    - Something new happened—need to react quickly!
- Finetuning can tricky
  - Training is sensitive to hyperparams
  - Not enough validation data
  - We don't quite understand how finetuning works
  - Expensive to train, time and memory

[ACL 2022 Tutorial Beltagy, Cohan, Logan IV, Min and Singh; quote credit: Colin Raffel]

In-Context Learning: Practically Useful

- Easier to access the “general knowledge” of pretrained model

Table 1 | Comparison among DeepSeek-V3.2-Base, DeepSeek-V4-Flash-Base, and DeepSeek-V4-Pro-Base. All models are evaluated in our internal framework and share the same evaluation setting. Scores with a gap not exceeding 0.3 are considered to be at the same level. The highest score in each row is in **bold font**, and the second is underlined.

	Benchmark (Metric)	# Shots	DeepSeek-V3.2 Base	DeepSeek-V4-Flash Base	DeepSeek-V4-Pro Base
	Architecture	-	MoE	MoE	MoE
	# Activated Params	-	37B	13B	49B
	# Total Params	-	671B	284B	1.6T
World Knowl.	AGIEval (EM)	0-shot	80.1	82.6	**83.1**
	MMLU (EM)	5-shot	87.8	88.7	**90.1**
	MMLU-Redux (EM)	5-shot	87.5	89.4	**90.8**
	MMLU-Pro (EM)	5-shot	65.5	68.3	**73.5**
	MMMLU (EM)	5-shot	87.9	88.8	**90.3**
	C-Eval (EM)	5-shot	90.4	92.1	**93.1**
	CMMLU (EM)	5-shot	88.9	90.4	**90.8**
	MultiLoKo (EM)	5-shot	38.7	42.2	**51.1**
	Simple-QA verified (EM)	25-shot	28.3	30.1	**55.2**
	SuperGPQA (EM)	5-shot	45.0	46.5	**53.9**
	FACTS Parametric (EM)	25-shot	27.1	33.9	**62.6**
	TriviaQA (EM)	5-shot	83.3	82.8	**85.6**
Lang. & Reas.	BBH (EM)	3-shot	**87.6**	86.9	**87.5**
	DROP (F1)	1-shot	88.2	**88.6**	**88.7**
	HellaSwag (EM)	0-shot	86.4	85.7	**88.0**
	WinoGrande (EM)	0-shot	78.9	79.5	**81.5**
	CLUEWSC (EM)	5-shot	83.5	82.2	**85.2**
Code & Math	BigCodeBench (Pass@1)	3-shot	**63.9**	56.8	59.2
	HumanEval (Pass@1)	0-shot	62.8	69.5	**76.8**
	GSM8K (EM)	8-shot	91.1	90.8	**92.6**
	MATH (EM)	4-shot	60.5	57.4	**64.5**
	MGSM (EM)	8-shot	81.3	**85.7**	84.4
	CMath (EM)	3-shot	92.6	**93.6**	90.9
Long Context	LongBench-V2 (EM)	1-shot	40.2	44.7	**51.5**

- DeepSeek’s Tech. Report 2026

In-Context Learning: Practically Useful



In-Context Learning: Skild's LocoFormer

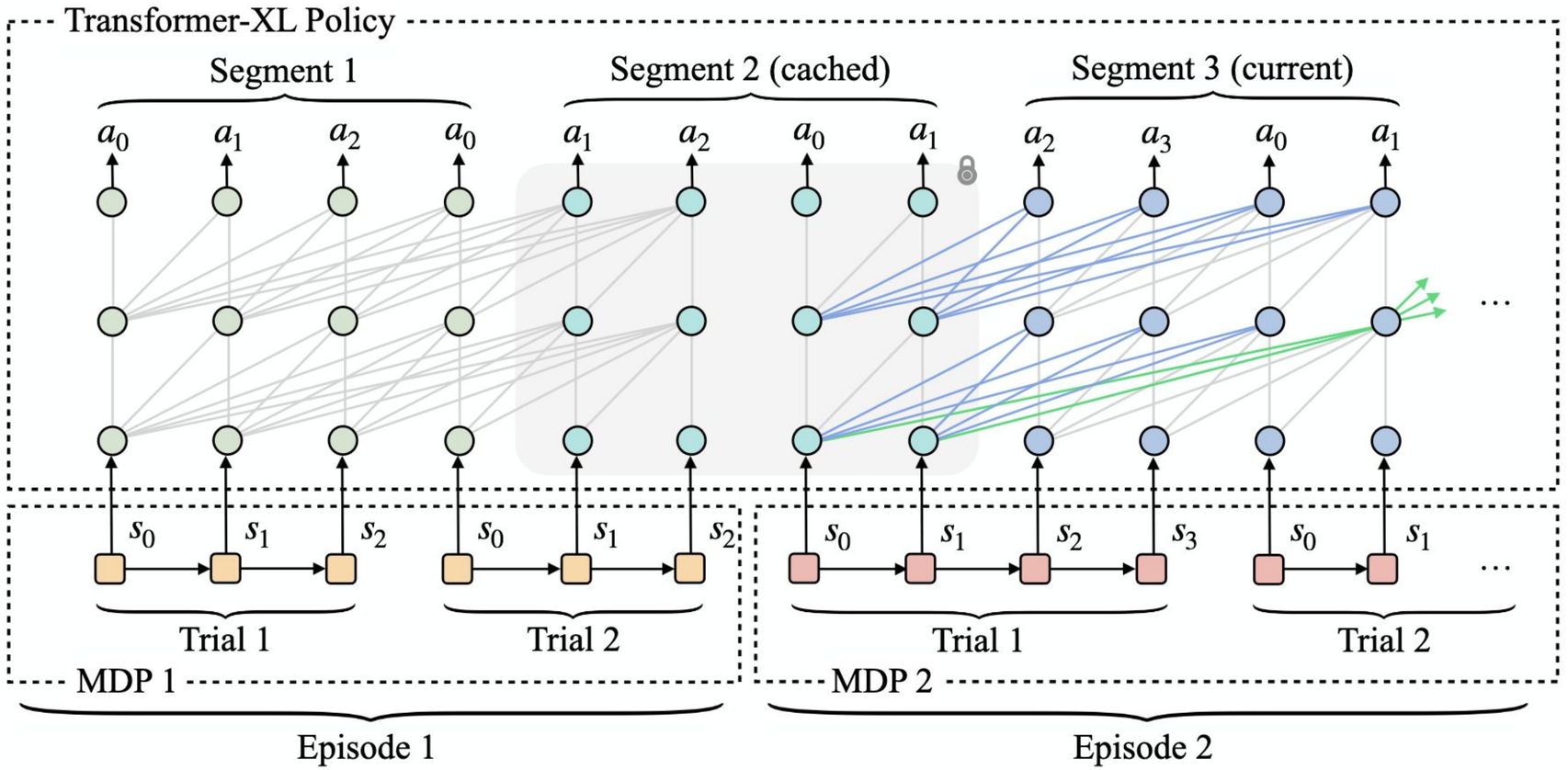


Figure 3: To enable long-context adaptation, we allow LocoFormer to attend to states from prior trials. We use a long-context Transformer backbone, that divides context into fixed-length segments. When processing the current segment, keys and values are also computed from states in the previous (cached) segment as well, but gradients are not propagated (blue lines). As the number of layers increases, this allows LocoFormer to use information even segments prior to the cached one (green lines).

LocoFormer: Generalist Locomotion via Long-context Adaptation. Min Liu et. al. 2025

In-Context Learning: Intellectually Intriguing

- Potential test for “Intelligent Behavior”
  - Generalization from few examples
    - Fundamental piece of intelligence
    - Often used in psychology
    - Quickly adjust to environment
- Insights into Language Modeling
  - What does an LLM “know”?
  - What are the biases/limitations of LLMs?
  - …

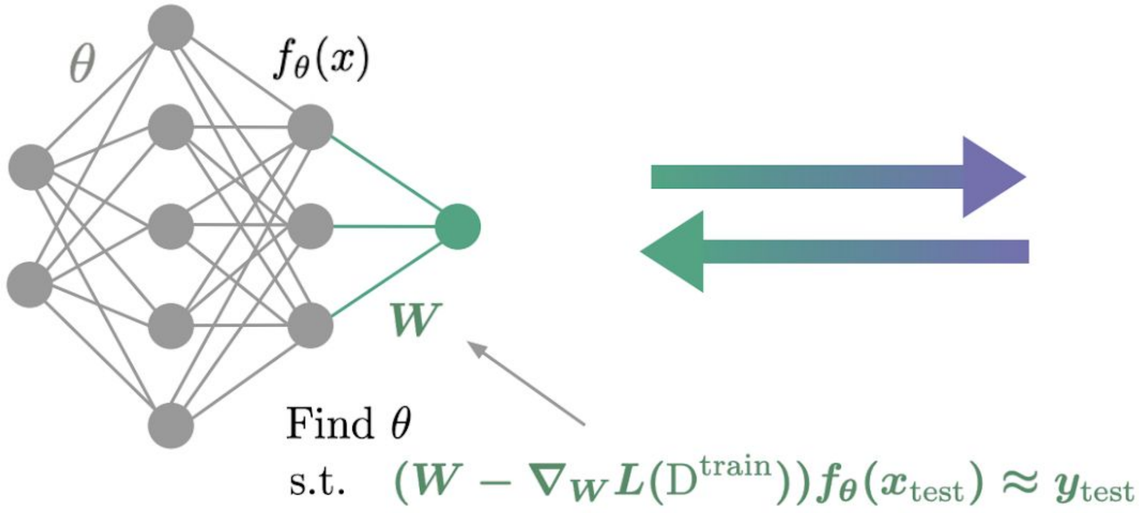
[ACL 2022 Tutorial Beltagy, Cohan, Logan IV, Min and Singh]

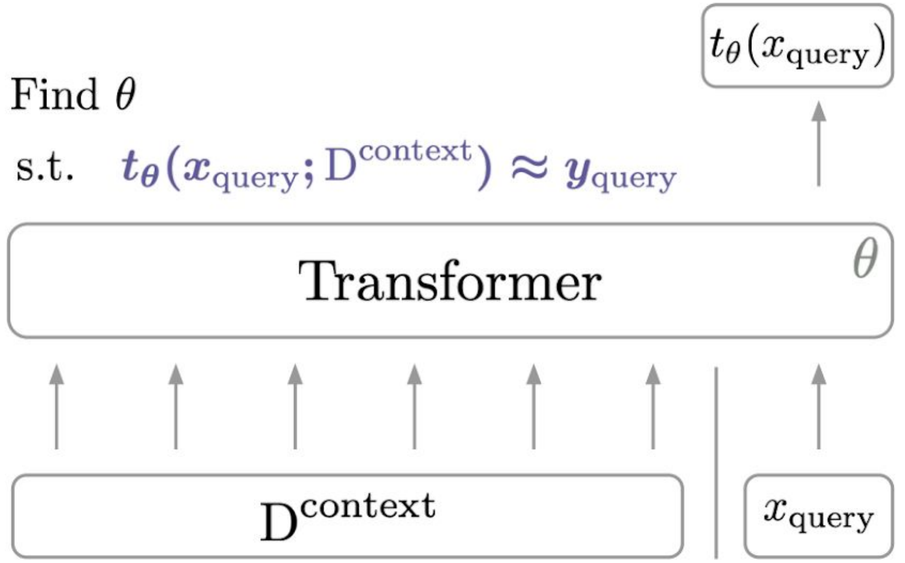
In-context Learning as Gradient Descent?

In-context Learning as Gradient Descent?

[von Oswald et al. 2022; Akyurek et al. 2022; Dai et al. 2022, ...]

- Hypothesis: ICL is implicitly equivalent to SGD on in-context demonstrations

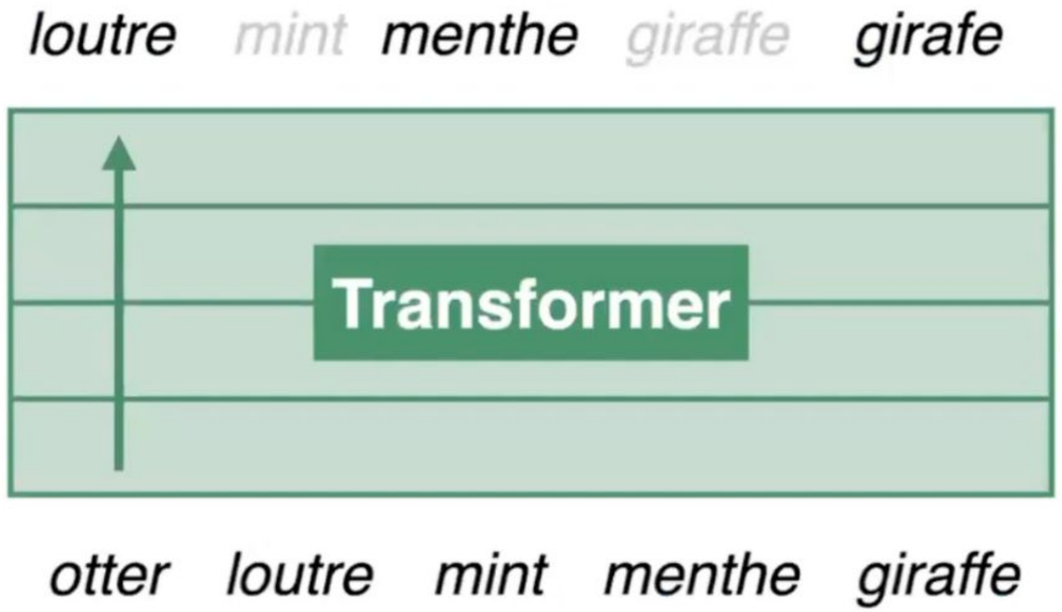




[Do Pre-trained Transformers Really Learn In-context by Gradient Descent?, 2024]

ICL for translation

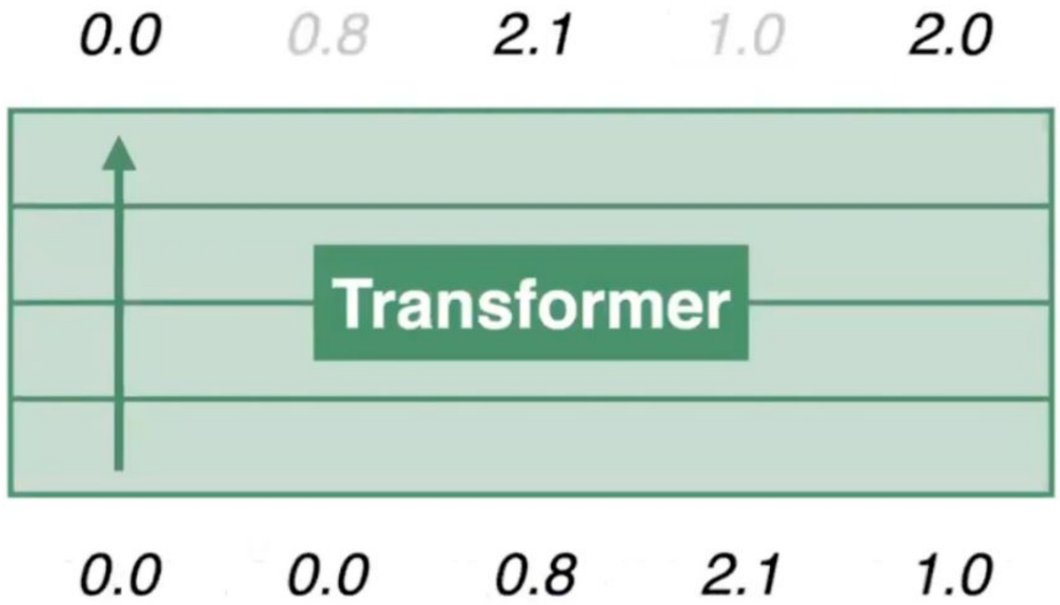
x	y
otter	*loutre*
mint	*menthe*
giraffe	*???*



ICL for regression

x	y
0.0	0.0
0.8	2.1
1.0	???

$$y = w^\top x$$



OLS regression

x	y
0.0	0.0
0.8	2.1
1.0	???

$$y = w^\top x$$

0.0 0.8 2.1 1.0 2.0

$$\hat{w} = (X^\top X)^{-1} X^\top f(x)$$

$$\hat{f}(x) = \hat{w}^\top x_n$$

0.0 0.0 0.8 2.1 1.0

Transformer layer and OLS

The Transformer should solve: $f_t(x_{1:t}) = \hat{W}_t x_t = X_t X_{t-1}^T (X_{t-1} X_{t-1}^T + \lambda I)^{-1} x_t$

We provide **weight constructions** for the following (after copying):

1) One causal *linear* self-attention **can implement a step of GD for *t* internal optimization problems** i.e. $f_t(x_{1:t}) = \hat{W}_t x_t = \eta X_t X_{t-1}^T x_t$ in parallel

2) One causal linear self-attention can compute the next summand of ***t*** *projected* **truncated Neumann series in parallel**

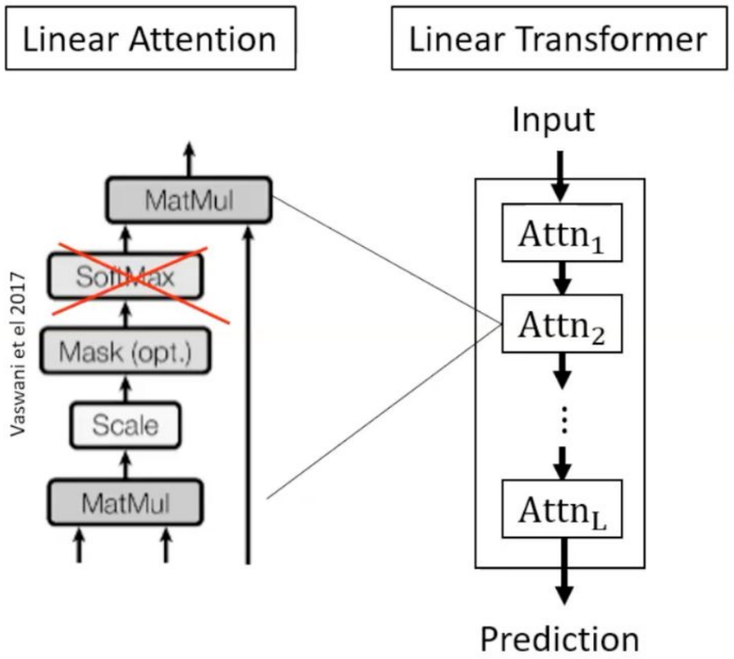
$$(X_{t-1} X_{t-1}^T + \lambda I)^{-1} x_t = \sum_{k=0}^{\infty} ((1-\lambda) I - X_{t-1} X_{t-1}^T)^k x_t$$

This implies that in **log(log(1/epsilon))-layers** the Transformer can approximate ***t*** matrix inverses to epsilon error.

Transformers learn in-context by gradient descent, von Oswald et. al, ICML, 2023

How does linear transformer in ICL?

remove softmax activation



- For **a single layer linear** transformer trained on the random instances of linear regression (esp. when the in-context samples are non-isotropic), the set of global minimizers corresponds to implementing one step of a (preconditioned) gradient descent on the least square objective. Here the least square objective is based on the examples given in the prompt.

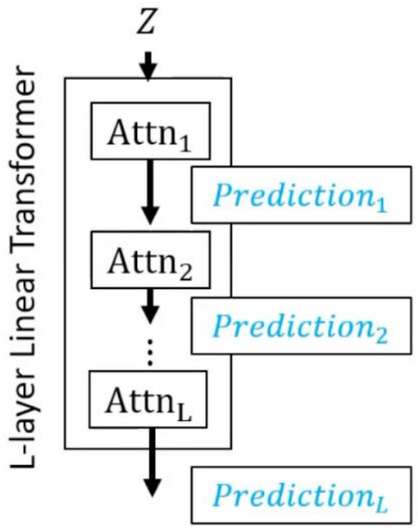
- For a **L-layer linear transforme**r trained on the random instances of linear regression, a subset of stationary points corresponds to implementing L steps of a (preconditioned) gradient descent on the least square objective.

Transformer learns to implement gradient descent

Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \cdots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & \cdots & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1)\times(n+1)}$

demonstrations | query

Goal: predict $y^{(n+1)}$



$Prediction_k$: The Transformer's prediction for $y^{(n+1)}$ at layer k, given input Z

Training objective:

$$\mathbb{E}_{(Z,\theta^*)}\left[\left(Prediction_L - y^{(n+1)}\right)^2\right]$$

(True label)

Transformer learns to implement gradient descent

Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \cdots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1)\times(n+1)}$

demonstrations | query

Goal: predict $y^{(n+1)}$

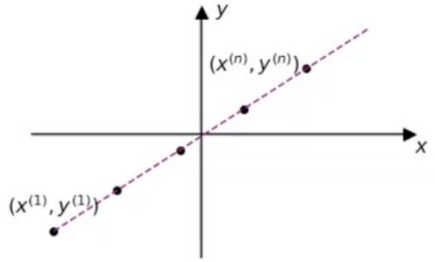
Training objective:

$\mathbb{E}\left[\left(Prediction_L - y^{(n+1)}\right)^2\right]$

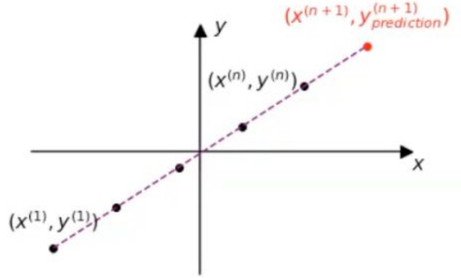
(True label)

Given n demonstrations, how do we predict $y^{(n+1)}$?

Step 1: Find best-fit $\hat{\theta}$ for demonstrations.



Step 2: Use $\hat{\theta}$ to predict $y^{(n+1)}$



Transformer learns to implement gradient descent

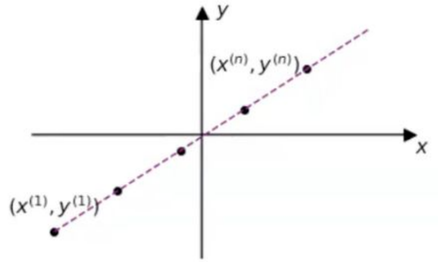
Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \cdots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1)\times(n+1)}$

demonstrations | query

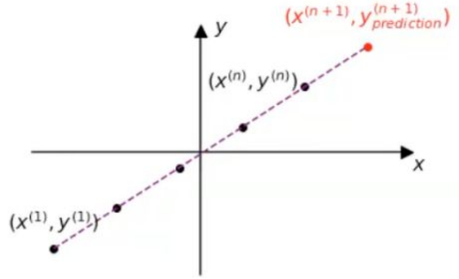
Goal: predict $y^{(n+1)}$

Given n demonstrations, how do we predict $y^{(n+1)}$?

Step 1: Find best-fit $\hat{\theta}$ for demonstrations.



Step 2: Use $\hat{\theta}$ to predict $y^{(n+1)}$



Training objective:

$\mathbb{E}\left[\left(Prediction_L - y^{(n+1)}\right)^2\right]$

(True label)

$\hat{\theta}$ minimizes the Empirical Loss

$R(\theta) = \sum_{i=1}^{n}\left(y^{(i)} - \theta^\top x^{(i)}\right)^2$

sum over n demonstrations

Transformer learns to implement gradient descent

Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \cdots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1)\times(n+1)}$

demonstrations | query

Goal: predict $y^{(n+1)}$

$$\theta_0^{GD} = 0$$

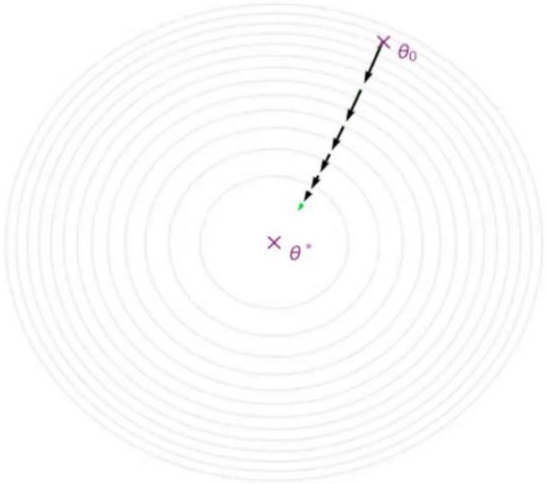
$$\theta_1^{GD} = \theta_0^{GD} - \delta_1 \nabla R\left(\theta_0^{GD}\right)$$

$$\theta_2^{GD} = \theta_1^{GD} - \delta_2 \nabla R\left(\theta_1^{GD}\right)$$

$$\vdots$$

$$\theta_L^{GD} = \theta_{L-1}^{GD} - \delta_L \nabla R\left(\theta_{L-1}^{GD}\right)$$

Gradient Descent on $R(\theta)$



Training objective:

$\mathbb{E}\left[\left(Prediction_L - y^{(n+1)}\right)^2\right]$

(True label)

$\hat{\theta}$ minimizes the Empirical Loss

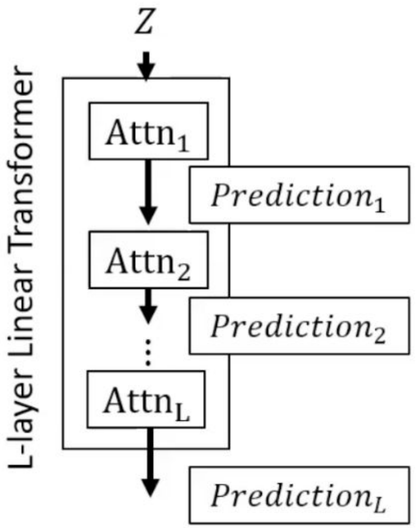
$R(\theta) = \sum_{i=1}^{n}\left(y^{(i)} - \theta^\top x^{(i)}\right)^2$

Transformer learns to implement gradient descent

Input: $Z = \begin{bmatrix} x^{(1)} & x^{(2)} & \cdots & x^{(n)} & x^{(n+1)} \\ y^{(1)} & y^{(2)} & \cdots & y^{(n)} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1)\times(n+1)}$

demonstrations | query

Goal: predict $y^{(n+1)}$



$\theta_0^{GD} = 0$

$\theta_1^{GD} = \theta_0^{GD} - \delta_1 \nabla R\left(\theta_0^{GD}\right)$

$\theta_2^{GD} = \theta_1^{GD} - \delta_2 \nabla R\left(\theta_1^{GD}\right)$

$\vdots$

$\theta_L^{GD} = \theta_{L-1}^{GD} - \delta_L \nabla R\left(\theta_{L-1}^{GD}\right)$

Gradient Descent on $R(\theta)$

Theorem 3 [ACDS 23]
Assume $x^{(i)} \sim \mathcal{N}(0, I)$, $\theta^* \sim \mathcal{N}(0, I)$.
There **exist stationary points** of the training objective, such that the Transformer prediction at layer k is the same as k steps of gradient descent on $R(\theta)$, i.e.

$Prediction_k = \langle \theta_k^{GD}, x^{(n+1)} \rangle$,

for $k = 1 \ldots L$

Training objective:
$\mathbb{E}\left[\left(Prediction_L - y^{(n+1)}\right)^2\right]$
(True label)

$\hat{\theta}$ minimizes the Empirical Loss
$R(\theta) = \sum_{i=1}^{n} \left(y^{(i)} - \theta^\top x^{(i)}\right)^2$

Theorem 1 [ACDS 23]
If layer = 1, the global minimum of the training objective coincides with one step of gradient descent.

Brief history

+: Formalize the in-context learning problem
+: Experiments: Transformers can be trained to achieve good ICL loss
- : Consider simple functions

Language Models are Few Shot Learners
Brown, Mann, Ryder, Subbiah, et el 2020

What Can Transformers Learn In-Context? A Case Study of Simple Function Classes
Garg, Tsipras, Liang, Valiant 2022

+: Propose mechanism for ICL via OLS
+: Experimental evidence by comparing loss over different tasks
- : Complicated construction, difficult to learn in practice
- : Consider simple functions

What learning algorithm is in-context learning? Investigations with linear models
Akyurek, Schuurmans, Andreas, Ma, Zhou 2023
Transformers as statisticians: Provable in-context learning with in-context algorithm selection
Lin, Bai, Mei 2023

+: Detailed mechanism – ICL via OLS via Gradient Descent
+: Some experimental evidence that proposed parameters are being learned
- : Linear Transformer applied to learning linear functions

Linear Transformers Are Secretly Fast Weight Programmers
Schlag, Irie, Schmidhuber 2021
Transformers Learn In-Context by Gradient Descent
Von Oswald, Niklasson, Randazzo, Sacramento, Mordvintsev, Zhmoginov, Vladymyrov 2023

+: Theoretical proof that Transformer implement GD at local/global minimum
+: Experimental evidence that proposed parameters are exactly being learned
- : Linear Transformer applied to learning linear functions

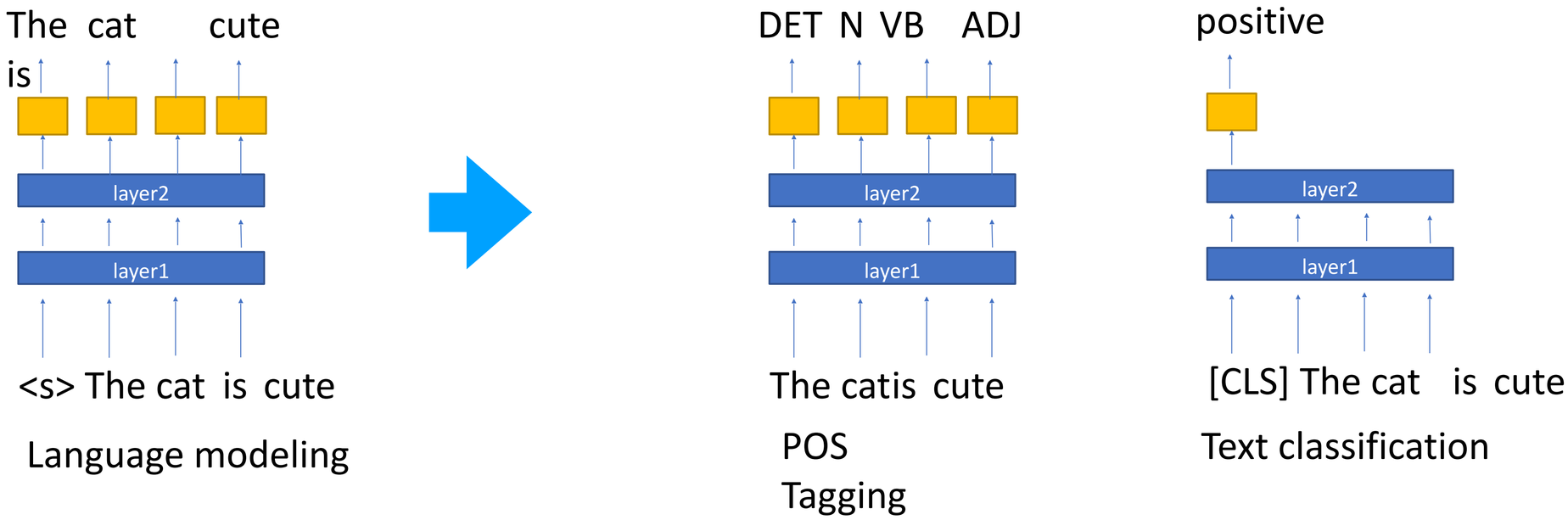
Transformers learn to implement preconditioned gradient descent for in-context learning
Ahn, Cheng, Daneshmand, Sra 2023
Trained transformers learn linear models in-context
R Zhang, S Frei, PL Bartlett
One step of gradient descent is provably the optimal in-context learner with one layer of linear self-attention
Mahankali, Hashimoto, Ma 2023

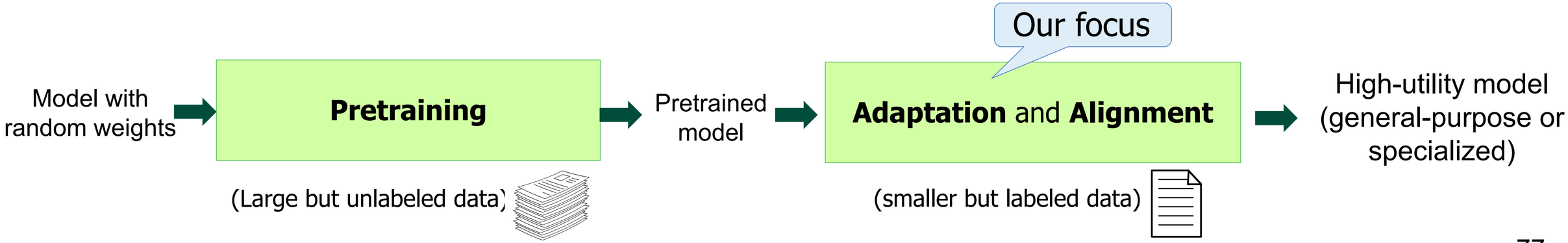
+: Nonlinear Transformer applied to Nonlinear Functions
+: ICL via functional gradient descent, related to choice of nonlinear activation
- : Several restrictions apply

Transformers implement functional gradient descent to learn non-linear functions in context
Cheng, Chen, Sra, 2024

Instruction-tuning

Fine-Tuning for Tasks





Language Modeling ≠ Following Human Instructions

PROMPT	*Explain the moon landing to a 6 year old in a few sentences.*
COMPLETION	GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
```

There is a mismatch between LLM pre-training and user intents.

Language Modeling ≠ Following Human Instructions

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION Human

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

There is a mismatch between LLM pre-training and user intents.

Language Modeling ≠ Incorporating Human Values

PROMPT *It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*

COMPLETION GPT-3
Adam

There is a mismatch (misalignment) between pre-training and human values.

Ethical-Advice Taker: Do Language Models Understand Natural Language Interventions?, Zhao et al. 2021

Language Modeling ≠ Incorporating Human Values

PROMPT *It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*

COMPLETION Human

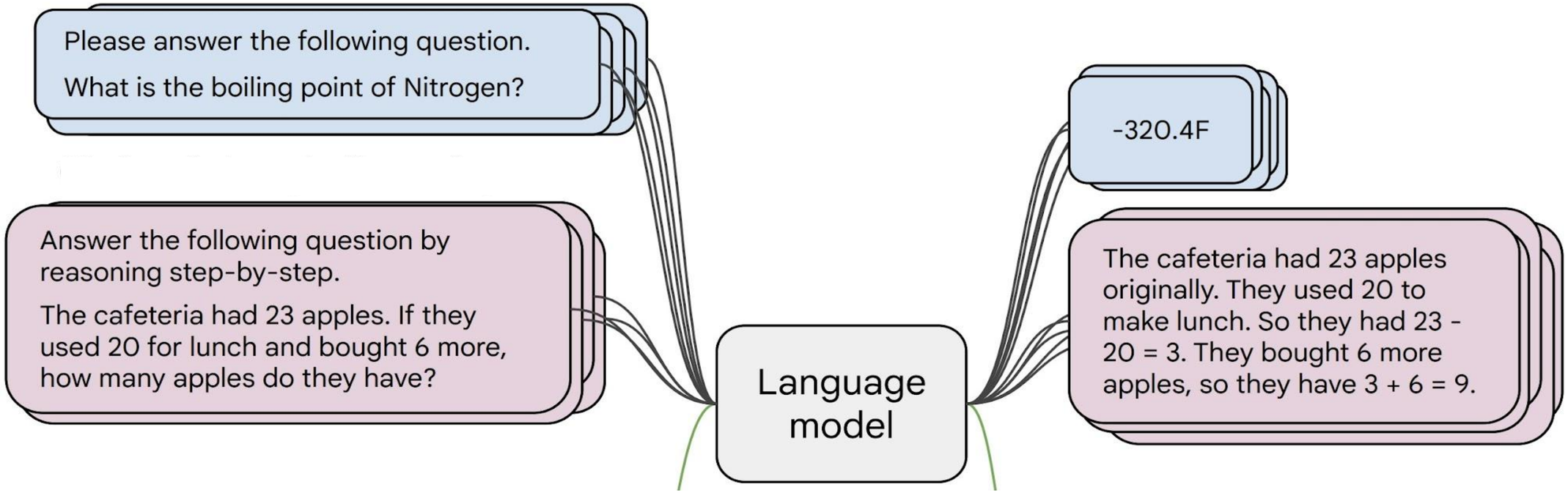
neither as we don't know much about their background or experience.

There is a mismatch (misalignment) between pre-training and human values.

Ethical-Advice Taker: Do Language Models Understand Natural Language Interventions?, Zhao et al. 2021

Instruction-tuning

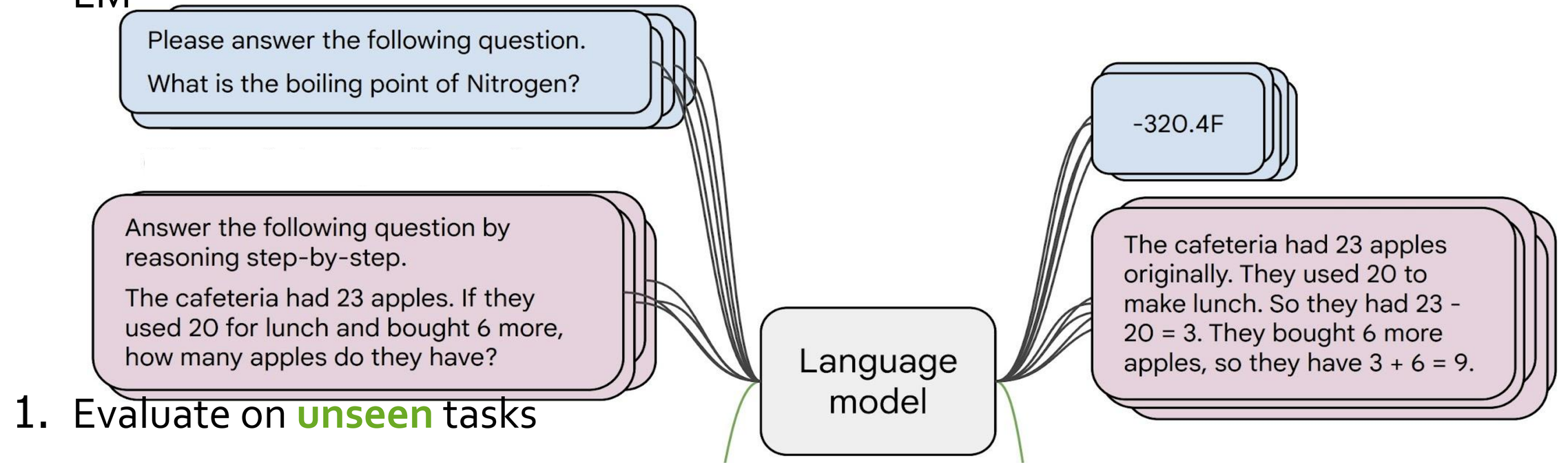
- Finetuning pre-trained LMs to map instructions to their corresponding responses.



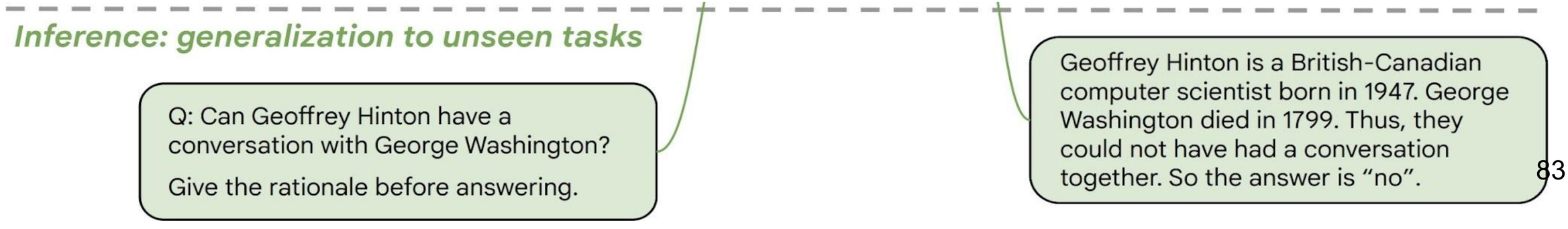
Instruction-tuning

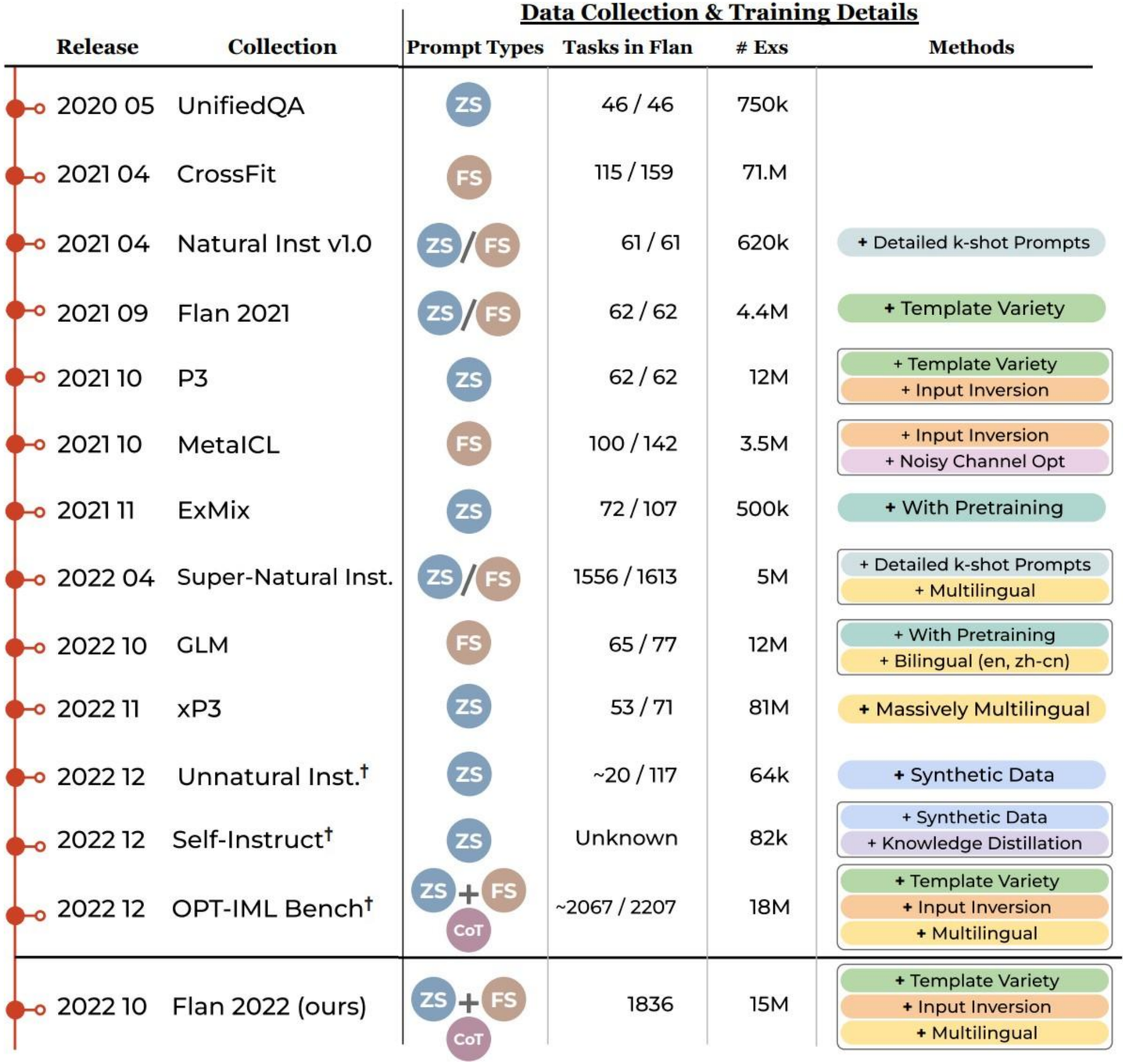
[Weller et al. 2020. Mishra et al. 2021; Wang et al. 2022, Sanh et al. 2022; Wei et al., 2022, Chung et al. 2022, many others]

1. Collect examples of (instruction, output) pairs across many tasks and finetune an LM



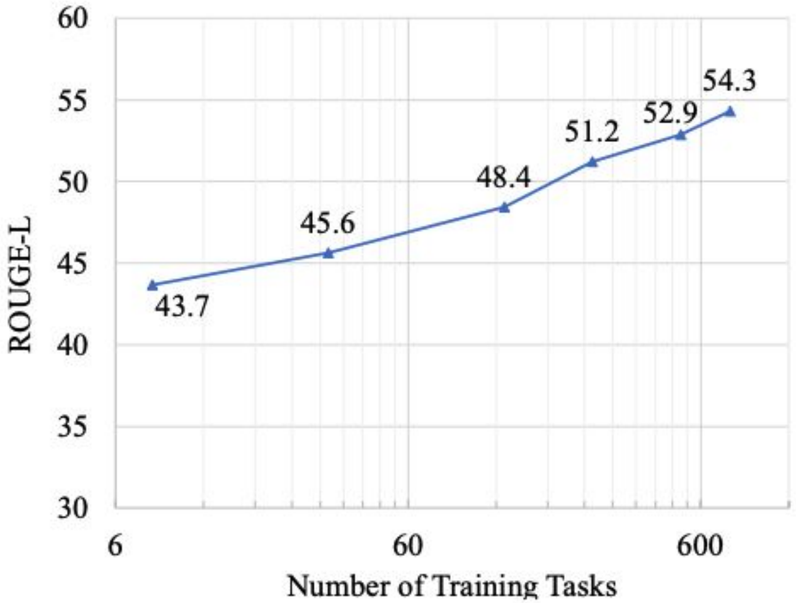
1. Evaluate on **unseen** tasks

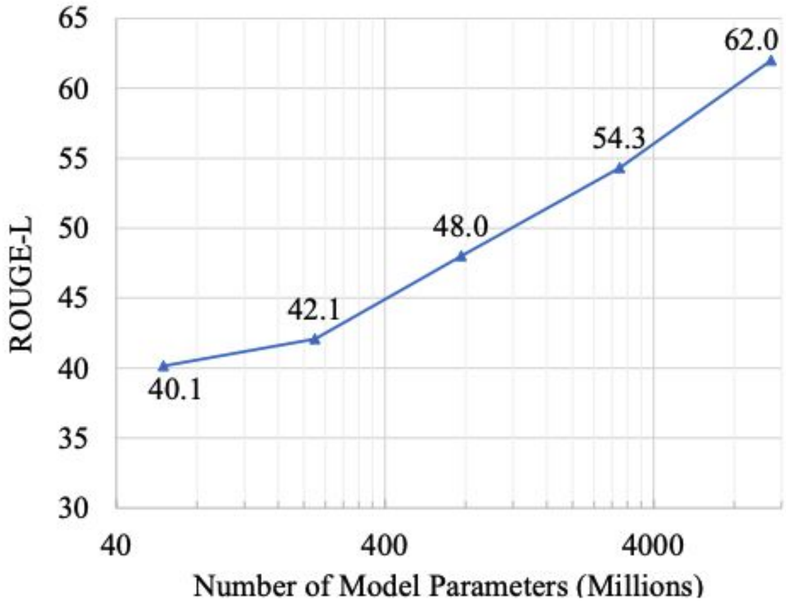




Designing Data and Methods for Effective Instruction Tuning (Longpre et al., 2023)

Scaling Instruction-Tuning



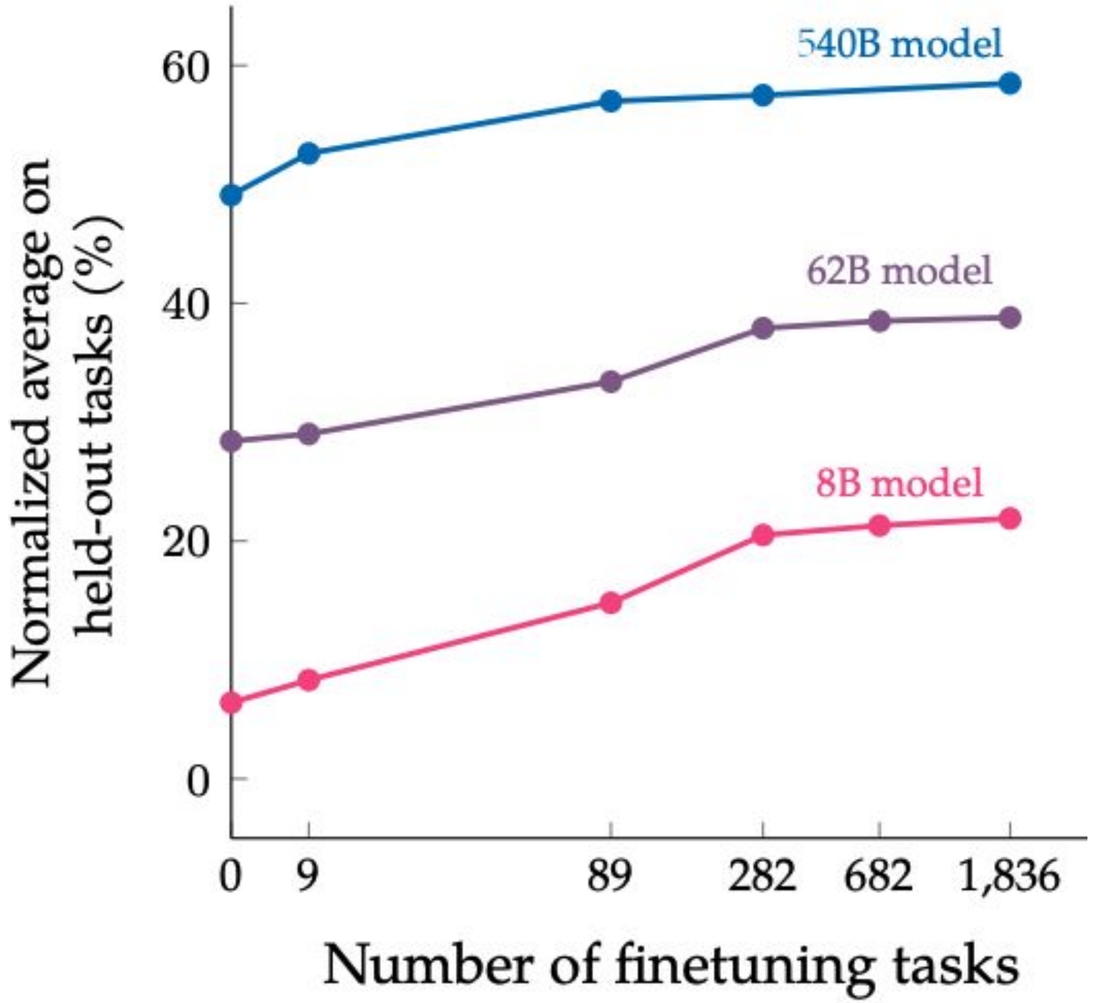


Linear growth of model performance
with exponential increase in observed tasks and model size.

Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ NLP Tasks, Wang et al. 2022
Cross-Task Generalization via Natural Language Crowdsourcing Instructions (Mishra et al., 2022)

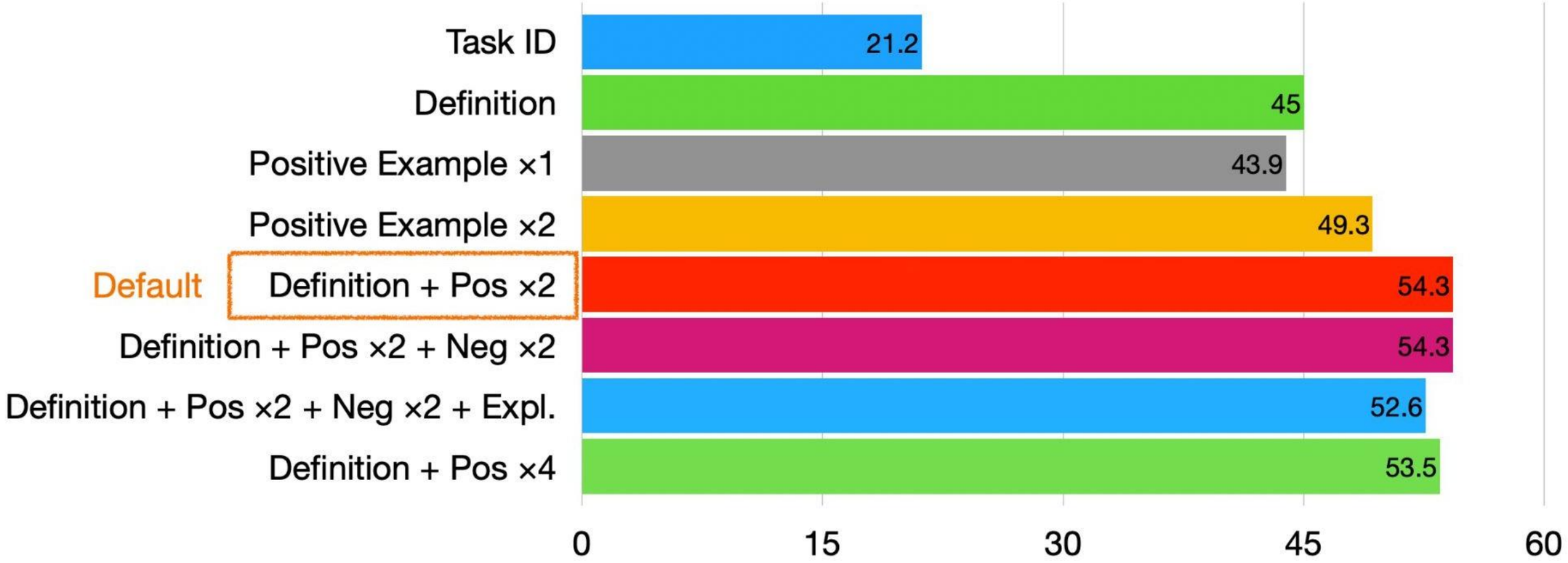
Scaling Instruction-Tuning

- Instruction finetuning improves performance by a large margin compared to no finetuning
- Increasing the number of finetuning tasks improves performance
- Increasing model scale by an order of magnitude (i.e., 8B → 62B or 62B → 540B) improves performance substantially for both finetuned and non-finetuned models



[Scaling Instruction-Finetuned Language Models, Chung et al. 2022]

Good instructions or in-context examples



Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ NLP Tasks, Wang et al. 2022

Cross-Task Generalization via Natural Language Crowdsourcing Instructions (Mishra et al., 2022)

Super-NI Takeaways

- Cross-task generalization via instructions is plausible.
- Super-NaturalInstructions provides a rich playground for such study.
- For instruction tuning:
  - Task/Instruction diversity is important!
  - Larger models bring in consistent improvement - not converged yet.
  - Large number of training instances could lead to overfitting to the training task.

Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ NLP Tasks, Wang et al. 2022
Cross-Task Generalization via Natural Language Crowdsourcing Instructions (Mishra et al., 2022)

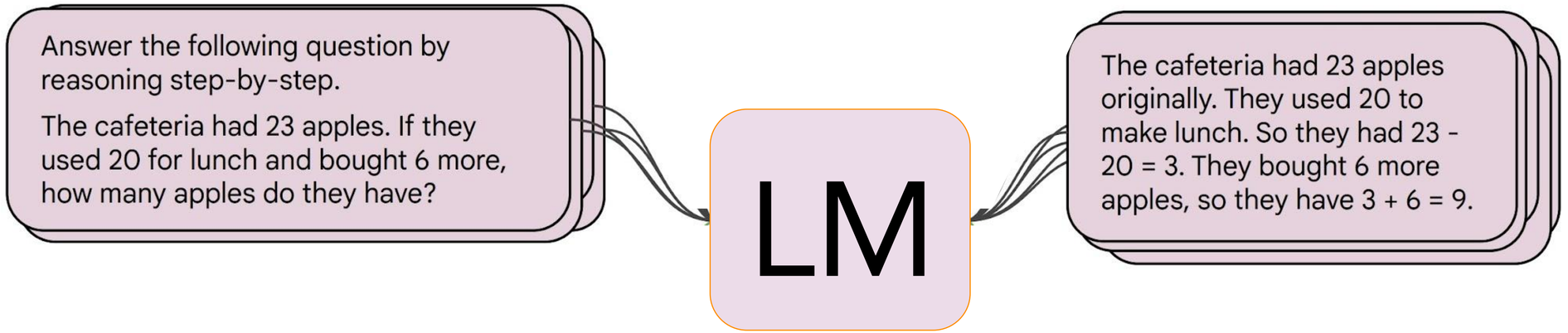
Instruction tuning doesn't have significant cost compared with pretraining

Params	Model	Architecture	Pre-training Objective	Pre-train FLOPs	Finetune FLOPs	% Finetune Compute
80M	Flan-T5-Small	encoder-decoder	span corruption	1.8E+20	2.9E+18	1.6%
250M	Flan-T5-Base	encoder-decoder	span corruption	6.6E+20	9.1E+18	1.4%
780M	Flan-T5-Large	encoder-decoder	span corruption	2.3E+21	2.4E+19	1.1%
3B	Flan-T5-XL	encoder-decoder	span corruption	9.0E+21	5.6E+19	0.6%
11B	Flan-T5-XXL	encoder-decoder	span corruption	3.3E+22	7.6E+19	0.2%
8B	Flan-PaLM	decoder-only	causal LM	3.7E+22	1.6E+20	0.4%
62B	Flan-PaLM	decoder-only	causal LM	2.9E+23	1.2E+21	0.4%
540B	Flan-PaLM	decoder-only	causal LM	2.5E+24	5.6E+21	0.2%
62B	Flan-cont-PaLM	decoder-only	causal LM	4.8E+23	1.8E+21	0.4%
540B	Flan-U-PaLM	decoder-only	prefix LM + span corruption	2.5E+23	5.6E+21	0.2%

Scaling Instruction-Finetuned Language Models (2022)

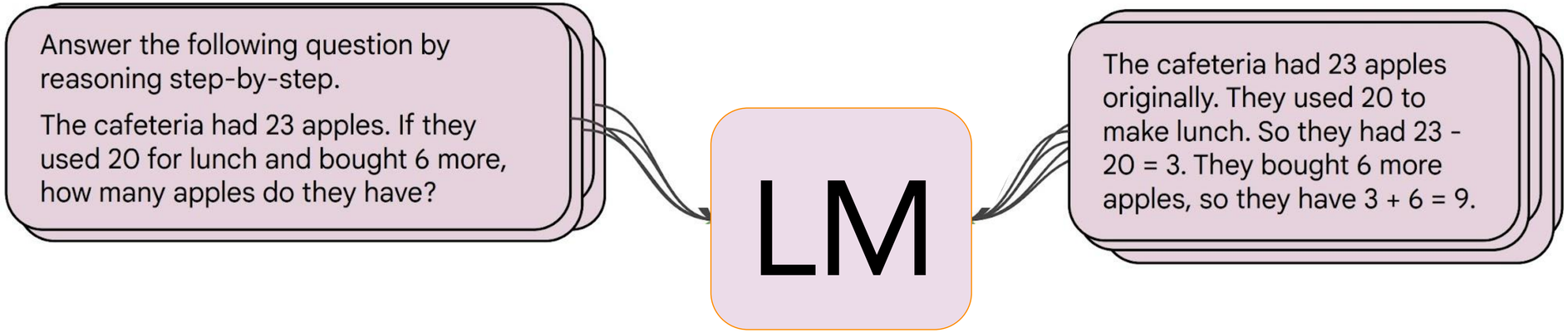
Instruction tuning

- Here is the recipe:
  - Prepare the data: diverse annotated data (instructions desired responses)
  - Split along tasks to train and test
  - Train on data of all training tasks:
    - › Optimize the per-token likelihood of the target (desired) responses
  - Test: zero-shot on new tasks



Limits of Instruction-Tuning

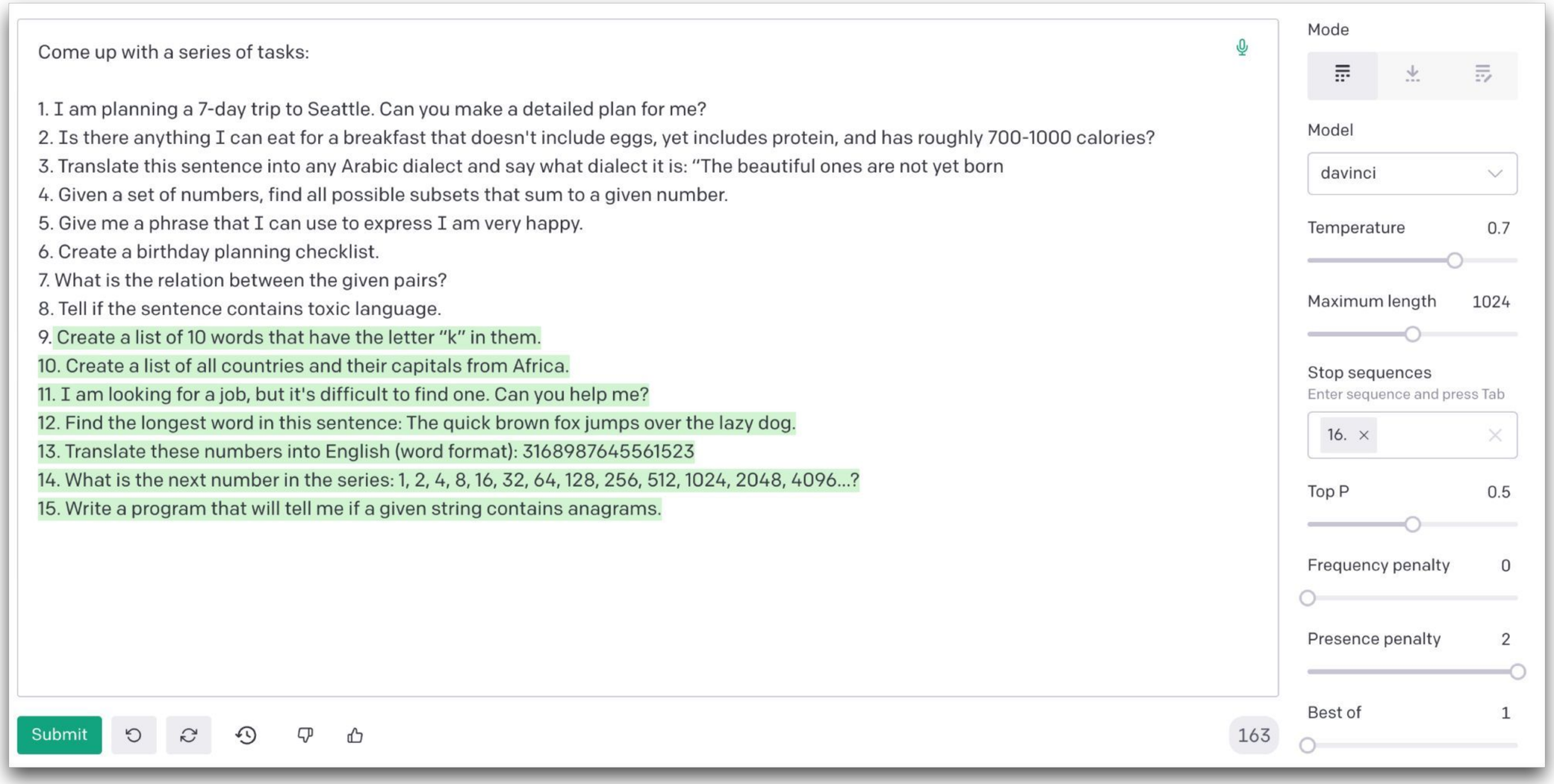
1. Difficult to collect diverse data.
2. Resulting models may not be good at open-ended generation tasks.
   - Incentivizes word-by-word rote learning => The resulting LM's generality/creativity is bounded by that of their supervision data.



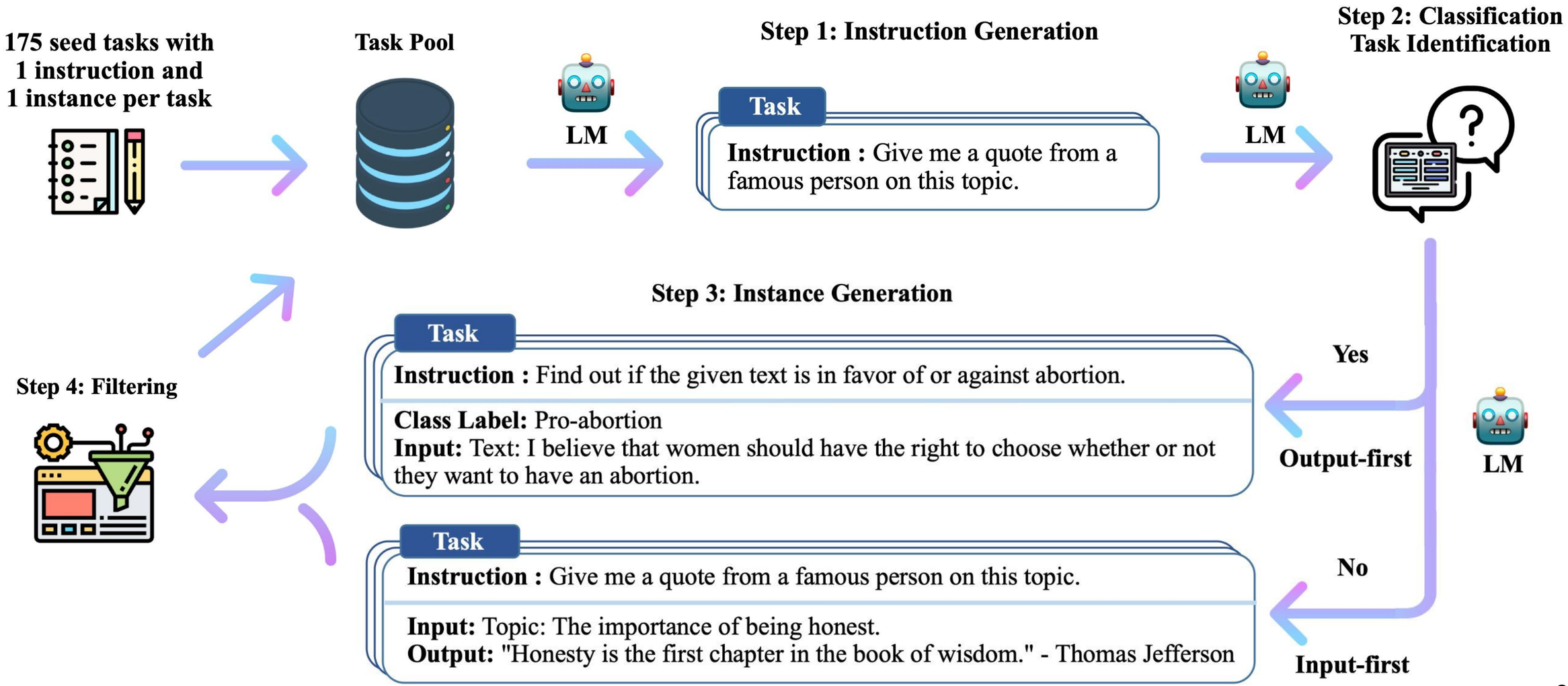
Limits of Instruction-Tuning

1. Difficult to collect diverse data.
2. Resulting models may not be good at open-ended generation tasks.
   - Incentivizes word-by-word rote learning => The resulting LM's generality/creativity is bounded by that of their supervision data.
3. Resulting models may hallucinate more regularly.
   - Labeled data is collected agnostic to the LM's knowledge => there might be a mismatch between labeled data and LM knowledge.
   - Hence, we may be encouraging "hypocritic" behavior => further hallucinations

LLMs can be prompted to generate instructions



Instruction data generation pipeline



Source: Orca: Progressive Learning from Complex Explanation Traces of GPT-4

Summary: Instruction fine-tuning (SFT)

SFT: Training LMs with annotated input instructions and their output.

Improves performance of LM's zero-shot ability in following instructions.

Factors: Data size, data diversity, model size

Works works best when we are just extracting pre-training behaviors, not adding new ones

Cons:

- It's expensive to collect ground- truth data for tasks.
- This is particularly difficult for open-ended creative generation have no right answer.
- Prone to hallucinations.